

Chapter 1 - Mathematical Models

1.1

Write the differential equation for a diffusion problem where the diffusing element decays at a rate proportional to the concentration.

Solution:

The diffusion equation in one dimension is

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} + \gamma u = f$$

where u is the concentration of the element and f is the source. The model for this problem is

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} = -\gamma u$$

Writing this out in three dimensions, it instead becomes

$$\frac{\partial u}{\partial t} - D \Delta u = -\gamma u$$

where $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

□

1.3

The temperature distribution in a thin isolated rod is $\frac{T_0}{1+x^2}$ at a specific moment in time. Calculate the heat flux. In what way is it directed?

Solution:

The heat flux is

$$\mathbf{j} = -\lambda \nabla T = \frac{\lambda 2x T_0}{(1+x^2)^2} \hat{x}$$

The direction is $\hat{x}$

□

1.4

Determine a differential equation for heat conduction in a thin rod, not perfectly isolated. The heat loss at the point x is proportional to the difference between the rod temperature and its surrounding temperature.

Solution:

$$\frac{1}{a} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = -\frac{c}{\lambda} (u - T_0)$$

□

1.5

Determine a differential equation for transversal vertical oscillations of a long horizontal string

- (a) under the influence of gravity,
- (b) if the surrounding medium additionally exerts a resistance proportional to the velocity.

Solution:

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f$$

- (a) Here, f is simply the gravitational force $F = mg$

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l}$$

where ρ_l is the length density.

- (b) Add a proportionality factor of $\frac{\partial u}{\partial x}$ which is the velocity

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l} - a \frac{\partial u}{\partial x}$$

□

1.8

Let $u(x, y)$ be the temperature in a plate Ω .

(a) How do you formulate a mathematical heat conduction problem with homogenous Neumann boundary conditions?

(b) What physical phenomenon gives rise to homogenous Neumann boundary conditions? Motivate.

In the figures below, the solutions to the problem is shown. Along some boundaries there are homogenous Neumann conditions, and along some there are homogenous Dirichlet conditions.

(c) Left figure shows the vector field, ∇u . Along which parts of the boundary is there homogenous Neumann conditions?

(d) Right figure shows the isotherms ($u = \text{constant}$). Along which parts of the boundary is there homogenous Neumann conditions?

Solution:

(a)

$$\frac{1}{a} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f$$

$$\frac{\partial u}{\partial x} = 0, \text{ on some boundary}$$

$$\frac{\partial u}{\partial y} = 0, \text{ on some boundary}$$

(b) Heat flux on boundary

(c) Top, top right, bottom left.

(d) Left, bottom and top right. □

1.10**Solution:**

□

1.11**Solution:**

□

1.12

A wide and tall wall with thickness L starts with temperature T_0 , same as surroundings. At time $t = 0$ the temperature on one side changes to T_1 . At the boundaries on the wall the coefficients are α_0 and α_1 . Determine the differential equation, initial values and boundary conditions.

Solution:

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0$$

$$u(L, t) = T_0$$

$$\frac{\partial u(0, t)}{\partial x} = \alpha_0(u(0, t) - T_1)$$

$$\frac{\partial u(L, t)}{\partial x} = -\alpha_1(u(L, t) - T_0)$$

□

1.13

Interpret the following in physical meaning for heat conduction.

(a) $u(0, t) = 0$

(b) $\frac{\partial u(0, t)}{\partial x} = 0$

(c) $\frac{\partial u(0, t)}{\partial x} - hu(0, t) = 0, \frac{\partial u(L, t)}{\partial x} + hu(L, t) = 0$

Solution:

- (a) Temperature at $x = 0$ always 0.
 - (b) No heat flux at $x = 0$, isolated.
 - (c) Heat flux on both boundaries, proportional to the difference in temperature between the medium and the surroundings. ☐
-

1.14**Solution:**☐**1.15****Solution:**☐**1.18****Solution:**☐**1.21****Solution:**☐

1.22

Solution:

☐

1.23

Solution:

☐

1.26

Solution:

☐

Chapter 3.1–3.3 - Fourier's Method I

3.2

Determine the coefficients in the following expansions

(a)

$$x = \sum_0^{\infty} \alpha_k \cos(k\pi x), \quad 0 < x < 1$$

(b)

$$x = \sum_1^{\infty} \alpha_k \sin(k\pi x), \quad 0 < x < 1$$

Solution:

(a)

$$\alpha_k = 2 \int_0^1 x \cos(k\pi x) \, dx = 2 \underbrace{\left[\frac{-x \sin(k\pi x)}{k\pi} \right]_0^1}_{=0} + 2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx =$$

$$= \frac{2}{k\pi} \int_0^1 \sin(k\pi x) \, dx = \frac{2}{k\pi} \left[\frac{\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} =$$

$$= \frac{2}{k^2 \pi^2} (\cos(k\pi) - 1) = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

$$\alpha_0 = \int_0^1 x \, dx = \frac{1}{2}$$

$$\alpha_k = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

↓

$$x = \frac{1}{2} + \frac{2}{\pi^2} \sum_0^{\infty} \frac{(-1)^k - 1}{k^2} \cos(k\pi x)$$

(b)

$$\begin{aligned}
 \beta_k &= 2 \int_0^1 x \sin(k\pi x) \, dx = 2 \left[\frac{x \cos(k\pi x)}{k\pi} \right]_0^1 - \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} \, dx}_{=0} = \\
 &= \frac{2 \cos(k\pi)}{k\pi} \\
 \beta_0 &= \int_0^1 x \sin(0\pi x) \, dx = 0 \\
 &\quad \downarrow \\
 x &= \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \sin(k\pi x)
 \end{aligned}$$

□

3A.5

Solve the heat problem

$$\begin{aligned}
 \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\
 u(0, t) = u(1, t) &= 0, & t > 0 \\
 u(x, 0) &= \sin(\pi x) + 2 \sin(3\pi x), & 0 < x < 1
 \end{aligned}$$

Solution:

We have homogenous Dirichlet conditions, which should give our solution in the form of a sinus series. We can therefore express our solution in an infinite sum

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin\left(\frac{k\pi x}{L}\right)$$

with $L = 1$. Using this expression we can differentiate termwise and put into the differential equation, resulting in

$$\begin{aligned}
& \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 \\
& \downarrow \\
& \sum_{k=1}^{\infty} \underbrace{(u'_k(t) + k^2 \pi^2 u_{k(t)})}_{=0} \sin(k\pi x) = 0 \\
& \downarrow \\
& u'_k(t) + k^2 \pi^2 u_{k(t)} = 0 \\
& \downarrow \\
& u_k(t) = c_k e^{-k^2 \pi^2 t}
\end{aligned}$$

This gives our function $u(x, t)$ as

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and using our initial conditions for $u(x, 0)$, we can determine the coefficients c_k :

$$\begin{aligned}
u(x, 0) &= \sum_{k=1}^{\infty} c_k \sin(k\pi x) = \sin(\pi x) + 2 \sin(3\pi x) \\
&\downarrow \\
c_k &= \begin{cases} 1, & k = 1 \\ 2, & k = 3 \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

This gives the final solution

$$u(x, t) = e^{-\pi^2 t} \sin(\pi x) + 2e^{-9\pi^2 t} \sin(3\pi x)$$

□

3A.7

A thin rod with closed ends has length $L = 1$. In the rod is air mixed with gas. Initially, the gas concentration is q in the left half of the rod, and 0 in the right half of the rod. Diffusion constant is set to $D = 1$. There is no source inside the rod. Write down a mathematical model and solve the problem.

Solution:

Using the diffusion equation

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(1, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= \begin{cases} q, & 0 < x < \frac{1}{2} \\ 0, & \frac{1}{2} < x < 1 \end{cases}\end{aligned}$$

We have homogenous Neumann conditions, which will give the solution on the form of a cosine series expansion.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) &= 0 \\ \downarrow \\ u'_{k(t)} + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t}\end{aligned}$$

This gives the solution on the form

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

and we only need to determine the coefficients c_k , which is done by the initial value $u(x, 0)$.

We want to expand the function $u(x, 0)$ as a cosine series

$$\begin{aligned}u(x, 0) &= \sum_{k=0}^{\infty} c_k \cos(k\pi x) \\ \downarrow \\ c_k &= 2 \int_0^{\frac{1}{2}} q \cos(k\pi x) \, dx = 2 \left[\frac{q \sin(k\pi x)}{k\pi} \right]_0^{\frac{1}{2}} = \frac{2q \sin(\frac{k\pi}{2})}{k\pi} \\ c_0 &= \int_0^{\frac{1}{2}} q \, dx = \frac{q}{2}\end{aligned}$$

Final solution is then

$$u(x, t) = \frac{q}{2} + \frac{2q}{\pi} \sum_{k=1}^{\infty} \frac{\sin\left(\frac{k\pi}{2}\right)}{k} e^{-k^2 \pi^2 t} \cos(\pi k x)$$

□

3A.8

Solve the heat problem

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\ u(x, 0) &= x, \quad 0 < x < 1 \end{aligned}$$

with the boundary conditions

(a)

$$u(0, t) = u(1, t) = 0, \quad t > 0$$

(b)

$$\frac{\partial u(0, t)}{\partial x} = \frac{\partial u(1, t)}{\partial x} = 0, \quad t > 0$$

Solution:

(a) This problem has homogenous Dirichlet conditions, which will give the solution on sinus form.

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=1}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \sin(k\pi x) &= 0 \\ \downarrow \\ u'_k(t) + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t} \end{aligned}$$

This gives

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and

$$u(x, 0) = x = \sum_{k=1}^{\infty} c_k \sin(k\pi x)$$

from which we can determine the coefficients c_k .

$$c_k = 2 \int_0^1 x \sin(k\pi x) dx = 2 \left[\frac{-x \cos(k\pi x)}{k\pi} \right]_0^1 + \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} dx}_{=0}$$

$$c_k = \frac{-2 \cos(k\pi)}{k\pi} = \frac{2}{k\pi} (-1)^{k+1}$$

$$c_0 = \int_0^1 x \sin(k\pi 0) dx = 0$$

Final solution is

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} e^{-k^2 \pi^2 t} \sin(k\pi x)$$

(b) This problem has homogenous Neumann conditions, which will give the solution on cosine form.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting into differential equation

$$\sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) = 0$$

↓

$$u'_k(t) + k^2 \pi^2 u_k(t) = 0$$

↓

$$u_k(t) = c_k e^{-k^2 \pi^2 t}$$

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

$$u(x, 0) = x = \sum_{k=0}^{\infty} c_k \cos(k\pi x)$$

From this we can determine the coefficients c_k

$$\begin{aligned}
c_k &= 2 \int_0^1 x \cos(k\pi x) \, dx = -2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx = \\
&= 2 \left[\frac{\cos(k\pi x)}{k^2 \pi^2} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} = \frac{2}{k^2 \pi^2} ((-1)^k - 1) \\
c_0 &= \int_0^1 x \cos(k\pi 0) \, dx = \frac{1}{2}
\end{aligned}$$

This gives our final solution as

$$u(x, t) = \frac{1}{2} + \frac{2}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k - 1}{k^2} e^{-k^2 \pi^2 t} \cos(k\pi x)$$

□

3A.10

Solution:

□

3A.12

Solve the oscillation problem

$$\begin{aligned}
\frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\
u(0, t) = u(1, t) &= 0, \quad t > 0 \\
u(x, 0) &= 1, \quad 0 < x < 1 \\
\frac{\partial u(x, 0)}{\partial t} &= 0, \quad 0 < x < 1
\end{aligned}$$

Solution:

We have homogenous Dirichlet conditions which will give the solution on sinus form

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Insert into differential equation and differentiate termwise

$$\sum_{k=1}^{\infty} (u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)) \sin(k\pi x) = 0$$

↓

$$u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)$$

Solving the characteristic equation with $u_k = re^t$

$$r^2 + 2r + k^2\pi^2 = 0$$

↓

$$r = -1 \pm \sqrt{1 - k^2\pi^2} = -1 \pm i\sqrt{k^2\pi^2 - 1} = -1 \pm i\omega_k$$

↓

$$u_k(t) = e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t))$$

Gives the solution as

$$u(x, t) = \sum_{k=1}^{\infty} e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) \sin(k\pi x)$$

$$u(x, 0) = 1 = \sum_{k=1}^{\infty} a_k \sin(k\pi x)$$

↓

$$a_k = 2 \int_0^1 \sin(k\pi x) dx = 2 \left[\frac{-\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{-2 \cos(k\pi)}{k\pi} + \frac{2}{k\pi} \rightarrow a_k = \frac{2}{k\pi} (1 - (-1)^k)$$

$$\frac{\partial u(x, 0)}{\partial t} = 0 =$$

$$= \sum_{k=1}^{\infty} (-e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) + e^{-t}(-a_k \omega_k \sin(\omega_k t) + b_k \omega_k \cos(\omega_k t))) \sin(k\pi x) =$$

$$= \sum_{k=1}^{\infty} (-a_k + \omega_k b_k) \sin(k\pi x) = 0$$

↓

$$-a_k + \omega_k b_k = 0 \rightarrow b_k = \frac{a_k}{\omega_k} = \frac{2}{\omega_k k\pi} (1 - (-1)^k)$$

Full solution:

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} e^{-t} \frac{1 - (-1)^k}{k} \left(\cos(\omega_k t) + \frac{\sin(\omega_k t)}{\omega_k} \right) \sin(k\pi x)$$

□

3A.13

Solution:

☐

3A.14

Solution:

☐

3A.15

Solution:

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3A.17

Solution:

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3A.19

Solution:

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3A.21

Solution:

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3A.22

Solution:

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3A.24a

Solution:

☐

3A.24bc

Solution:

☐

3A.25

Solution:

☐

Chapter 3.4–3.5 - Fourier's Method I

3B.29

A test of water taken in a thin long cylindrical container of length L . At the start, the container only has water. At $t = 0$, one end of the cylinder is opened, where the substance concentration is q_0 . Solve the diffusion problem.

Solution:

Diffusion equation

$$\begin{aligned}\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < L, t > 0 \\ u(0, t) &= q_0, \quad t > 0 \\ \frac{\partial u(L, t)}{\partial x} &= 0, \quad t > 0 \\ u(x, 0) &= 0, \quad 0 < x < L\end{aligned}$$

We start by making the equations homogenous. Introduce

$$v(x, t) = u(x, t) - q_0$$

This gives the model in terms of $v(x)$:

$$\begin{aligned}\frac{\partial v}{\partial t} - D \frac{\partial^2 v}{\partial x^2} &= 0, \quad 0 < x < L, t > 0 \\ v(0, t) &= 0, \quad t > 0 \\ \frac{\partial v(L, t)}{\partial x} &= 0, \quad t > 0 \\ v(x, 0) &= -q_0, \quad 0 < x < L\end{aligned}$$

Expressed in terms of a linear operator, this becomes

$$\begin{aligned}\frac{\partial v}{\partial t} + D\mathcal{A}v &= 0, \quad D\mathcal{A} = \{v \in \mathcal{C}^2[0, L] \mid v(0) = 0, v'(L) = 0\} \\ \mathcal{A} &= -\frac{\partial^2}{\partial x^2}\end{aligned}$$

We now want to find the eigenvalues and eigenfunctions of the operator $\mathcal{A}$

$$\begin{aligned}
 \mathcal{A}\varphi = \lambda u &\rightarrow \frac{\partial^2 \varphi}{\partial x^2} + \lambda \varphi = 0 \rightarrow \varphi(x) = c_k e^{i\sqrt{\lambda_k}x} = \\
 &= A_k \cos(\sqrt{\lambda_k}x) + B_k \sin(\sqrt{\lambda_k}x) \\
 &\quad \downarrow \\
 \varphi(0) &= A_k \cos(\sqrt{\lambda_k}x) = 0 \rightarrow A_k = 0 \\
 \varphi'(L) &= \sqrt{\lambda_k} B_k \cos(\sqrt{\lambda_k}L) = 0 \rightarrow \sqrt{\lambda_k}L = \frac{\pi}{2} + k\pi
 \end{aligned}$$

This gives

$$\begin{aligned}
 \lambda_k &= \left(\frac{\pi}{L} \left[\frac{1}{2} + k \right] \right)^2 \\
 \varphi_k(x) &= \sin\left(\frac{\pi}{L} \left[\frac{1}{2} + k \right] x \right), \quad k = 0, 1, 2, \dots
 \end{aligned}$$

and we have found the eigenvalues and eigenfunctions of $\mathcal{A}$

We now have

$$v(x, t) = \sum_{k=0}^{\infty} v_k(t) \varphi_k(x)$$

Putting this into the differential equation for $v(x, t)$ gives

$$\begin{aligned}
 \sum_{k=0}^{\infty} (v'_k(t) + D\lambda_k v_k(t)) \varphi_k(x) &= 0 \\
 &\quad \downarrow \\
 v'_k(t) + D\lambda_k v_k(t) &= 0 \\
 &\quad \downarrow \\
 v_k(t) &= c_k e^{-D\lambda_k t}
 \end{aligned}$$

Now we need to find the coefficients c_k , which is done by calculating $v(x, 0)$:

$$v(x, 0) = \sum_{k=0}^{\infty} c_k e^{-D\lambda_k \cdot 0} \varphi_k(x) = \sum_{k=0}^{\infty} c_k \varphi_k(x) = -q_0$$

The coefficients are now found by using the projection formula

$$\begin{aligned}
 c_k &= \frac{(\varphi_k | -q_0)}{(\varphi_k | \varphi_k)} = \frac{-q_0 \int_0^L \sin(\sqrt{\lambda_k}x) dx}{\int_0^L \sin^2(\sqrt{\lambda_k}x) dx} = \\
 &= q_0 \frac{\left[\frac{\cos(\sqrt{\lambda_k}x)}{\sqrt{\lambda_k}} \right]_0^L}{\left[\frac{1}{2}x - \frac{\sin(2\sqrt{\lambda_k}x)}{4\sqrt{\lambda_k}} \right]_0^L} = q_0 \frac{\frac{\cos(\sqrt{\lambda_k}L)}{\sqrt{\lambda_k}} - \frac{1}{\sqrt{\lambda_k}}}{\frac{L}{2}} = -\frac{2q_0}{L\sqrt{\lambda_k}} = v_k(0)
 \end{aligned}$$

This gives

$$\begin{aligned}
 v_k(t) &= -\frac{2q_0}{L\sqrt{\lambda_k}} e^{-D\lambda_k t} \\
 &\downarrow \\
 v(x,t) &= -\frac{2q_0}{L} \sum_{k=0}^{\infty} \frac{1}{\sqrt{\lambda_k}} e^{-D\lambda_k t} \sin(\sqrt{\lambda_k} x) \\
 &\downarrow \\
 u(x,t) &= v(x,t) + q_0 = q_0 - \frac{2q_0}{L} \sum_{k=0}^{\infty} \frac{1}{\sqrt{\lambda_k}} e^{-D\lambda_k t} \sin(\sqrt{\lambda_k} x)
 \end{aligned}$$

□

3B.30

Solution:

□

3B.32a

Solution:

□

3B.35

Determine the modes and eigenfrequencies for sound waves in an exponential horn described by letting

$$y = ke^{ax}, \quad 0 \leq x \leq L$$

rotate about the x-axis. The horn is open in both ends. The wave equation for a variable area is given by

$$\frac{\partial^2 u}{\partial t^2} - v^2 \frac{1}{A(x)} \frac{\partial}{\partial x} \left(A(x) \frac{\partial u}{\partial x} \right) = 0$$

Solution:

The area of the horn is

$$A(x) = \pi y^2 = \pi k^2 e^{2ax}$$

The differential equation can be described by a linear Sturm-Liouville operator

$$\mathcal{A} = -\frac{1}{A(x)} \frac{\partial}{\partial x} \left(A(x) \frac{\partial \varphi}{\partial x} \right)$$

$$D_{\mathcal{A}} = \{ \varphi \in \mathcal{C}^2[0, L] \mid \varphi(0) = \varphi(L) = 0 \}$$

Find eigenvalues and eigenfunctions to the operator

$$\begin{aligned} \mathcal{A}\varphi = \lambda\varphi &\rightarrow -\frac{1}{A(x)} \frac{\partial}{\partial x} \left(A(x) \frac{\partial \varphi}{\partial x} \right) = \lambda\varphi(x) \\ &\downarrow \\ -\frac{1}{A(x)} [A'(x)\varphi'(x) + A(x)\varphi''(x)] &= \lambda\varphi(x) \\ &\downarrow \\ -2a\varphi'(x) - \varphi''(x) = \lambda\varphi(x) &\rightarrow \varphi''(x) + 2a\varphi'(x) + \lambda\varphi(x) \end{aligned}$$

This gives the characteristic equation

$$r^2 + 2ar + \lambda = 0 \rightarrow r = -a \pm \sqrt{a^2 - \lambda}$$

This gives three cases for the square root:

$$\begin{cases} (1): a^2 - \lambda > 0 \\ (2): a^2 - \lambda = 0 \\ (3): a^2 - \lambda < 0 \end{cases}$$

For the first case:

$$\begin{aligned} (1): r &= -a \pm \sqrt{a^2 - \lambda} \\ &\downarrow \\ \varphi(x) &= c_1 e^{(-a+\sqrt{a^2-\lambda})x} + c_2 e^{(-a-\sqrt{a^2-\lambda})x} \\ \varphi(0) &= c_1 + c_2 = 0 \rightarrow c_1 = -c_2 \\ \varphi(L) &= c_1 e^{(-a+\sqrt{a^2-\lambda})L} - c_1 e^{(-a-\sqrt{a^2-\lambda})L} \rightarrow e^{-\sqrt{a^2-\lambda}} - e^{\sqrt{a^2-\lambda}} = 0 \end{aligned}$$

This is not possible, hence $c_1 = c_2 = 0$ and this corresponds to no eigenvalues or eigenfunctions.

Case 2:

$$(2): r = -a$$

$$\downarrow$$

$$\varphi(x) = (c_1 + c_2 x)e^{-ax}$$

$$\varphi(0) = c_1 = 0 \rightarrow c_1 = 0$$

$$\varphi(L) = c_2 L e^{-aL} = 0 \rightarrow c_2 = 0$$

This also gives no non-trivial solution as both c_1 and c_2 are zero.

Case 3:

$$(3): r = -a \pm i\sqrt{\lambda_k - a^2} = -a \pm i\omega_k, \text{ with } \omega_k = \sqrt{\lambda_k - a^2}$$

$$\downarrow$$

$$\varphi(x) = e^{-ax} (A \cos(\omega_k x) + B \sin(\omega_k x))$$

$$\varphi(0) = A \cos(\omega_k 0) \rightarrow A = 0$$

$$\varphi(L) = e^{-aL} \sin(\omega_k L) = 0 \rightarrow \omega_k L = \sqrt{\lambda_k - a^2} = k\pi$$

This gives the eigenvalues and eigenfunctions

$$\lambda_k = \frac{k^2 \pi^2}{L^2} + a^2$$

$$\varphi_k(x) = e^{-ax} \sin\left(\frac{k\pi x}{L}\right), \quad k = 1, 2, 3, \dots$$

k cannot be zero because λ has to be larger than a^2 , and $k = 0$ gives $\lambda_0 = a^2$.

□

3B.39

Solution:

□

3B.41

Solution:

□

3B.42

Solution:

☐

3B.43+60

Solution:

☐

3B.45a

Solution:

☐

3B.46

Solution:

☐

3B.49a

Solution:

☐

3B.51

Solution:

☐

3B.55

Solution:

☐

3B.62

Solution:

☐

3B.66

Solution:

☐

3B.69

Solution:

☐

Chapter 4 - Fourier's Method II

4.1

Solve the heat conduction problem

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad t > 0, x \in \mathbb{R} \\ u(x, 0) &= e^{-x^2}, \quad x \in \mathbb{R}\end{aligned}$$

Solution:

The solution will take the form

$$u(x, t) = \int_{\mathbb{R}} T(\xi, t) e^{ix\xi} d\xi$$

We have the relation

$$2\pi T(\xi, t) = (\mathcal{F}_x u)(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx$$

where $\mathcal{F}_x u$ is the partial Fourier transform of u in x . $u(x, 0) = e^{-x^2}$ is bounded. Taking the partial Fourier transform of the differential equation yields

$$\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx - \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx = 0 \quad (1)$$

We now assume it's possible to swap integration and derivation

$$\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx = \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx = \frac{\partial \hat{u}}{\partial t}(\xi, t)$$

where

$$\hat{u}(\xi, t) = \mathcal{F}_x u(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx$$

Since the partial Fourier transform is done in x , the second term in Equation 1 can be rewritten with the Fourier transform rule for derivatives

$$\int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx = -\xi^2 \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx = -\xi^2 \hat{u}(\xi, t)$$

Putting this into the differential equation gives

$$\mathcal{F}_x \left(\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial \hat{u}}{\partial t}(\xi, t) + \xi^2 \hat{u}(\xi, t) = 0$$

This has the general solution

$$\hat{u}(\xi, t) = c(\xi) e^{-\xi^2 t}$$

To determine $c(\xi)$, we set $t = 0$

$$\begin{aligned} c(\xi) = \hat{u}(\xi, 0) &= \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} e^{-x^2} e^{-ix\xi} dx = \\ &= \mathcal{F} [e^{-x^2}] (\xi) = \sqrt{\pi} e^{-\frac{\xi^2}{4}} \end{aligned}$$

This gives

$$\hat{u}(\xi, t) = \sqrt{\pi} e^{(-\xi^2(t+\frac{1}{4}))} = \sqrt{\pi} e^{-\xi^2(t+\frac{1}{4})}$$

To get the solution we need to inverse transform this

$$\begin{aligned} \mathcal{F} [e^{-x^2}] &= \sqrt{\pi} e^{-\frac{\xi^2}{4}} \\ \mathcal{F} [e^{-(ax)^2}] &= \frac{1}{|a|} \sqrt{\pi} e^{-\frac{(\frac{\xi}{a})^2}{4}} \end{aligned}$$

From this we see we want $t + \frac{1}{4} = \frac{1}{4a^2} \rightarrow a = \sqrt{\frac{1}{4(t+\frac{1}{4})}} \rightarrow \frac{1}{a} = \sqrt{4t+1}$. With this, we get

$$\mathcal{F} \left[\frac{1}{\sqrt{4t+1}} e^{-\left(\frac{x}{\sqrt{4t+1}}\right)^2} \right] = \sqrt{\pi} e^{-\xi^2(t+\frac{1}{4})}$$

↓

$$u(x, t) = \frac{1}{\sqrt{4t+1}} e^{-\left(\frac{x}{\sqrt{4t+1}}\right)^2}$$

□

4.3

Solve the heat problem

$$\begin{aligned}
\frac{\partial u}{\partial t} - a \frac{\partial^2 u}{\partial x^2} &= -\alpha u, & t > 0, x \in \mathbb{R} \\
u(x, 0) &= \delta(x), & x \in \mathbb{R} \\
u(x, t) &\rightarrow 0, x \rightarrow \pm\infty, & t > 0
\end{aligned}$$

Solution:

This problem describes a heat problem where there is a sudden impulse of heat at $x = 0, t = 0$. Partial Fourier transforming gives

$$\begin{aligned}
&\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx - a \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx + \alpha \int_{-\infty}^{\infty} u e^{-ix\xi} dx \\
&\quad \downarrow \\
&\frac{\partial \hat{u}}{\partial t}(\xi, t) + a\xi^2 \hat{u}(\xi, t) + \alpha \hat{u}(\xi, t) = 0 \\
&\quad \downarrow \\
&\frac{\partial \hat{u}}{\partial t}(\xi, t) + (a\xi^2 + \alpha) \hat{u}(\xi, t) = 0 \\
&\quad \downarrow \\
&\hat{u}(\xi, t) = c(\xi) e^{-(a\xi^2 + \alpha)t}
\end{aligned}$$

To determine $c(\xi)$, set $t = 0$ in $\hat{u}(\xi, t)$

$$\hat{u}(\xi, 0) = \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \delta(x) e^{-ix\xi} dx = 1$$

This means

$$\hat{u}(\xi, t) = e^{-(a\xi^2 + \alpha)t} = e^{-a\xi^2 t} e^{-\alpha t}$$

Now $u(x, t)$ is given by the inverse transform of this

$$\begin{aligned}
\mathcal{F} [e^{-x^2}] &= \sqrt{\pi} e^{-\frac{\xi^2}{4}} \\
\mathcal{F} \left[\frac{e^{-\alpha t}}{\sqrt{\pi}} e^{-x^2} \right] &= e^{-\frac{\xi^2}{4}} e^{-\alpha t} \\
\mathcal{F} \left[\frac{e^{-\alpha t}}{2\sqrt{\pi at}} e^{-\left(\frac{x}{2\sqrt{a}}\right)^2} \right] &= e^{-\frac{(\xi 2\sqrt{at})^2}{4}} e^{-\alpha t}
\end{aligned}$$

Final answer

$$u(x, t) = \frac{e^{-\alpha t}}{\sqrt{4\pi at}} e^{-\frac{x^2}{4at}}$$

□

4.5

Find a solution to the Schrödinger equation

$$\frac{\partial \psi}{\partial t} - i \frac{\partial^2 \psi}{\partial x^2} = 0, \quad t > 0, x \in \mathbb{R}$$

that satisfy the initial value

$$\psi(x, 0) = \sin^2(x), \quad x \in \mathbb{R}$$

Solution:

We will partial Fourier transform the differential equation, which results in

$$\int_{-\infty}^{\infty} \frac{\partial \psi}{\partial t} e^{-ix\xi} dx + i\xi^2 \int_{-\infty}^{\infty} \psi e^{-ix\xi} dx = 0$$

$$\downarrow$$

$$\frac{\partial \hat{\psi}}{\partial t}(\xi, t) + i\xi^2 \hat{\psi}(\xi, t) = 0$$

$$\downarrow$$

$$\hat{\psi}(\xi, t) = c(\xi) e^{-i\xi^2 t}$$

To determine $c(\xi)$, we set $t = 0$ and get

$$\begin{aligned} \hat{\psi}(\xi, 0) = c(\xi) &= \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \sin^2(x) e^{-ix\xi} dx = \\ &= \int_{-\infty}^{\infty} \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^2 e^{-ix\xi} dx = -\frac{1}{4} \int_{-\infty}^{\infty} (e^{2ix} - 2 + e^{-2ix}) e^{-ix\xi} dx \end{aligned}$$

$$\downarrow$$

$$c(\xi) = -\frac{1}{4} \mathcal{F} [e^{2ix} - 2 + e^{-2ix}] = -\frac{1}{4} [2\pi\delta(\xi - 2) - 4\pi\delta(\xi) + 2\pi\delta(\xi + 2)] =$$

$$\rightarrow c(\xi) = \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)]$$

This gives our $\hat{\psi}(\xi, t)$

$$\begin{aligned} \hat{\psi}(\xi, t) &= \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)] e^{-i\xi^2 t} = \\ &= \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2) \end{aligned}$$

and our solution $\psi(x, t)$ is the inverse Fourier transform of $\hat{\psi}(\xi, t)$

$$\mathcal{F} \left[\frac{1}{2} - \frac{1}{4} e^{-4it} e^{2ix} - \frac{1}{4} e^{-4it} e^{-2ix} \right] = \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2)$$

So our solution that satisfy the initial conditions is

$$\psi(x, t) = \frac{1}{2} - \frac{1}{4}e^{-4it}e^{2ix} - \frac{1}{4}e^{-4it}e^{-2ix} = \frac{1}{2} \left(1 - e^{-4it} \left[\frac{e^{2ix} + e^{-2ix}}{2} \right] \right)$$

↓

$$\psi(x, t) = \frac{1}{2}(1 - e^{-4it} \cos(2x))$$

□

4.6

Solve the problem

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} &= 0, & t > 0, x \in \mathbb{R} \\ u(x, 0) &= 0, \frac{\partial u}{\partial t}(x, 0) = \delta(x), & x \in \mathbb{R} \end{aligned}$$

Solution:

Partial Fourier Transform:

Partial Fourier transform gives

$$\begin{aligned} \frac{\partial^2 \hat{u}}{\partial t^2}(\xi, t) + \xi^2 \hat{u}(\xi, t) &= 0 \\ \hat{u}(\xi, t) &= A(\xi) \cos(\xi t) + B(\xi) \sin(\xi t) \end{aligned}$$

To determine $A(\xi)$ and $B(\xi)$, we set $t = 0$.

$$\begin{aligned} \hat{u}(\xi, 0) &= \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = 0 \rightarrow A(\xi) = 0 \\ \frac{\partial \hat{u}}{\partial t}(\xi, 0) &= \int_{-\infty}^{\infty} \frac{\partial u}{\partial t}(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \delta(x) e^{-ix\xi} dx = 1 \\ \frac{\partial \hat{u}}{\partial t}(\xi, 0) &= \xi B(\xi) \cos(\xi \cdot 0) = \xi B(\xi) = 1 \rightarrow B(\xi) = \frac{1}{\xi} \end{aligned}$$

We get our $\hat{u}(\xi, t)$ as

$$\hat{u}(\xi, t) = \frac{\sin(\xi t)}{\xi}$$

which has the inverse Fourier transform

$$\mathcal{F}^{-1} \left[\frac{\sin(\xi t)}{\xi} \right] = \frac{1}{2}(\theta(x+t) - \theta(x-t))$$

Partial Laplace Transform:

We can also solve it using the partial Laplace transform, which might be more convenient for wave equations. We set

$$U = \int_{-\infty}^{\infty} u(x, t) e^{-sx} dx$$

as the Laplace transform. Laplace transforming the differential equation in x gives

$$\int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial t^2} e^{-sx} dx - \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-sx} dx = 0$$

↓

$$\frac{\partial^2 U}{\partial t^2} - s^2 U = 0$$

↓

$$U(s, t) = A(s)e^{st} + B(s)e^{-st}$$

To determine the constants $A(s)$ and $B(s)$, we set $t = 0$

$$U(s, 0) = A(s) + B(s) = \int_{-\infty}^{\infty} u(x, 0) e^{-sx} dx = 0 \rightarrow A(s) = -B(s)$$

$$\frac{\partial U}{\partial t}(s, 0) = sA(s) - sB(s) = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t}(x, 0) e^{-sx} dx = \int_{-\infty}^{\infty} \delta(x) e^{-sx} dx = 1$$

↓

$$sA(s) - sB(s) = 1 \rightarrow 2sA(s) = 1 \rightarrow A(s) = \frac{1}{2s}, B(s) = -\frac{1}{2s}$$

↓

$$U(s, t) = \frac{1}{2s}(e^{st} - e^{-st})$$

Now all that's left to do is find the inverse Laplace transform of $U(s, t)$:

$$\mathcal{L} [\theta(x)] = \frac{1}{s}$$

$$\mathcal{L} [\theta(x - a)] = \frac{1}{s} e^{-as}$$

$$\mathcal{L} \left[\frac{1}{2}(\theta(x+t) - \theta(x-t)) \right] = \frac{1}{2s}(e^{st} - e^{-st})$$

↓

$$u(x, t) = \frac{1}{2}[\theta(x+t) - \theta(x-t)]$$

☐

4.7

Solution:

☐

4.8

Solution:

☐

4.10

Solution:

☐

4.14

Solution:

☐

4.15

Solution:

☐

4.16

Solution:

☐

4.17

Solution:

☐

4.20

Solution:

☐

Chapter 5 - Green's Functions

5.1

Solution:

☐

5.2

Solution:

☐

5.4

Solution:

☐

5.9

Solution:

☐

5.10

Solution:

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5.11

Solution:

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5.12

Solution:

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5.13

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5.20

Solution:

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5.22

Solution:

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5.24

Solution:

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5.27

Solution:

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5.28

Solution:

☐

Chapter 7 - Radiating Waves

7.1

Solution:

☐

7.2

Solution:

☐

7.3

Solution:

☐

7.4

Solution:

☐

7.6

Solution:

☐

7.8ab

Solution:

☐

7.8c

Solution:

☐

7.9

Solution:

☐

7.10

Solution:

☐

7.18

Solution:

□

Distributions

D.1

Solution:

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} &= -\frac{m_0 g}{\rho} \delta(x-a), \quad 0 < x < L, t > 0 \\ u(0, t) &= u(L, t) = 0, \quad t > 0 \\ u(x, t) &= g(x), \quad \frac{\partial u(x, 0)}{\partial t} = h(x), \quad 0 < x < L\end{aligned}$$

□

D.4

- (a) $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$
 (b) $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$

Solution:

- (a) Here we use $g(x)\delta(x) = g(0)\delta(x)$, and start by differentiating the expression $(g(x)\delta(x))$:

$$\begin{aligned}(g(x)\delta(x))' &= g'(x)\delta(x) + g(x)\delta'(x) \\ &\downarrow \\ g(x)\delta'(x) &= (g(x)\delta(x))' - g'(x)\delta(x) = \\ &= (g(0)\delta(x))' - g'(0)\delta(x) = g(0)\delta'(x) - g'(0)\delta(x)\end{aligned}$$

- (b) Using the result from (a) and differentiating $(g(x)\delta'(x))$:

$$\begin{aligned}
(g(x)\delta'(x))' &= g'(x)\delta'(x) + g(x)\delta''(x) \\
&\downarrow \\
g(x)\delta''(x) &= (g(x)\delta'(x))' - g'(x)\delta'(x) = \\
&= (g(0)\delta'(x) - g'(0)\delta(x))' - g'(x)\delta'(x) = \\
&= g(0)\delta''(x) - g'(0)\delta'(x) - g'(0)\delta'(x) + g''(0)\delta(x) = \\
&= g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)
\end{aligned}$$

□

D.5

- (a) $e^x \delta_1$
- (b) $x\delta'$
- (c) $\cos(x)\delta'$
- (d) $x^2\delta''$
- (e) $\int g(t)\delta(t-1) dt$

Solution:

(a)

$$e^x \delta_1 = e^x \delta_1 = e \delta_1$$

(b) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

$$x\delta' = -\delta$$

(c) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$:

$$\cos(x)\delta' = \cos(0)\delta' + \sin(0)\delta = \delta'$$

(d) Using $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$:

$$x^2\delta'' = 2\delta$$

(e) Using $\int g(x)\delta(x-a) dx = g(a)$:

$$\int g(t)\delta(t-1)dt = g(1)$$

□

D.6

(a) $f(x) = e^{-x}\theta(x)$

(b) $f(x) = e^{-|x|}$

Solution:

(a) Using $\theta'(x) = \delta(x)$

$$f'(x) = (e^{-x}\theta(x))' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \delta(x) - e^{-x}\theta(x)$$

$$f''(x) = (\delta(x) - e^{-x}\theta(x))' = \delta'(x) + e^{-x}\theta(x) - e^{-x}\delta(x) = \delta'(x) - \delta(x) + e^{-x}\theta(x)$$

(b) Expressing $|x|$ in distributions:

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} = x\theta(x) - x(1 - \theta(x)) = 2x\theta(x) - x$$

$$f'(x) = (-2\theta(x) - 2x\delta(x) + 1)e^{-|x|} = (1 - 2\theta(x))e^{-|x|}$$

$$\begin{aligned} f''(x) &= (e^{-|x|} - 2e^{-|x|}\theta(x))' = (1 - 2\theta(x))e^{-|x|} + 2(1 - 2\theta(x))e^{-|x|}\theta(x) - 2e^{-|x|}\delta(x) = \\ &= e^{-|x|}[(1 - 2\theta(x)) + 2\theta(x)(1 - 2\theta(x))] - 2\delta(x) = e^{-|x|}(1 - 2\theta(x))^2 - 2\delta(x) = \\ &= [(1 - 2\theta(x))^2 = 1 \text{ for all } x \neq 0] = e^{-|x|} - 2\delta(x) \end{aligned}$$

□

D.8

(b) Determine all primitive distributions for $x\theta(x-1)$

Solution:

(a)

$$(x^2\theta(x-1))' = 2x\theta(x-1) + x^2\theta'(x-1) = 2x\theta(x-1) + \delta(x-1)$$

(b) Try to compensate for the $\delta(x-1)$ in (a) by $x^2 \rightarrow x^2 - 1$:

$$((x^2 - 1)\theta(x-1))' = 2x\theta(x-1) + (x^2 - 1)\delta(x-1) = 2x\theta(x-1)$$

Now compensate with a factor 2 and we get the solution

$$\left(\frac{x^2 - 1}{2}\right)\theta(x-1) + C$$

□

D.9

Solution:

$$\frac{\partial^2 u}{\partial t^2} - S \frac{\partial^2 u}{\partial x^2} = F\delta(x-a)$$

$$u(0, t) = u(1, t) = 0$$

Since we are looking for the stationary solution, $\frac{\partial^2 u}{\partial t^2} = 0$ and we have

$$-\frac{\partial^2 u}{\partial x^2} = \frac{F}{S}\delta(x-a)$$

Integrating both sides twice gives

$$-u(x, t) = \frac{F}{S}(x-a)\theta(x-a) + Ax + B$$

Now check the boundary conditions for this expression:

$$u(0, t) = \frac{F}{S}(-a)\theta(-a) + B = 0 \rightarrow B = 0$$

$\underbrace{\hspace{1.5cm}}_{=0 \text{ since } a>0}$

$$u(1, t) = \frac{F}{S}(a-1)\theta(1-a) + A = 0 \rightarrow A = \frac{F}{S}(1-a)$$

$$u(x, t) = \frac{F}{S}((x(1-a) + (a-x)\theta(x-a)))$$

□

D.10**Solution:**

□

D.12

$$u'(x) + (2x + 1)u(x) = \delta'(x)$$

Solution:

Integrating factor for LHS is

$$e^{x^2+x}$$

Multiply this into the equation and integrate LHS, then using the expansion for $f(x)\delta'(x)$:

$$\begin{aligned} \left(u(x)e^{x^2+x}\right)' &= e^{x^2+x}\delta'(x) = e^0\delta'(x) - (2x+1)_{x=0}\delta(x) = \\ &= \left(u(x)e^{x^2+x}\right)' = \delta'(x) - \delta(x) \end{aligned}$$

Now integrating both sides gives:

$$u(x)e^{x^2+x} = \delta(x) - \theta(x) + C$$

$$\downarrow$$

$$u(x) = e^{-(x^2+x)}(\delta(x) - \theta(x) + C) = \delta(x) + e^{-x^2-x}(C - \theta(x))$$

□

D.15

- (a) $xU = 0$
- (b) $(x-1)U = 0$
- (c) $(e^x - 2)U = 0$
- (d) $(x^2 - 1)U = 0$
- (e) $\sin(x)U = 0$
- (f) $x^3U = 0$

Solution:

(a)

$$U = \delta(x) \rightarrow x\delta(x) = 0 \rightarrow U = c\delta(x), \quad c = \text{constant}$$

(b) Same as in (a) but $\delta(x - 1)$

$$U = c\delta(x - 1)$$

(c) We can try to set U as $\delta(x - a)$ to make $(e^x - 2) = 0 \rightarrow e^x = 2 \rightarrow x = \ln(2)$

$$U = c\delta(x - \ln(2))$$

(d) Same idea as above, but here we have two roots, $x = 1, x = -1$

$$U = c_1\delta(x - 1) + c_2\delta(x + 1)$$

(e) Can use $U = \delta(x - a)$ again to make $\sin(x) = 0$

$$U = c_k\delta(x - k\pi) = \sum_{k=-\infty}^{\infty} c_k\delta(x - k\pi)$$

(f) Triple root at $x = 0$, use the relation $x^k U = 0 \rightarrow U = \sum_{k=0}^{n-1} c_k \delta^{(k)}(x)$

$$U = c_0\delta(x) + c_1\delta'(x) + c_2\delta''(x)$$

□

D.17(a) $\theta(-2x + 1)$ (b) $\delta(2x)$ (c) $\delta(-2x + 1)$ (d) $\delta'(2x)$ **Solution:**

(a)

$$\theta(-2x+1) = \begin{cases} 1, & -2x+1 > 0 \rightarrow x < \frac{1}{2} \\ 0, & -2x+1 < 0 \rightarrow x > \frac{1}{2} \end{cases}$$

↓

$$\theta(-2x+1) = \theta\left(-2\left(x - \frac{1}{2}\right)\right) = 1 - \theta\left(x - \frac{1}{2}\right) = 1 - \theta_{\frac{1}{2}}(x)$$

(b) Scaling rule gives

$$\delta(2x) = \frac{1}{2}\delta(x)$$

(c)

$$\delta(-x) = \delta(x), \delta(ax) = \frac{1}{|a|}\delta(x)$$

↓

$$\delta(-2x+1) = \delta\left(-2\left(x - \frac{1}{2}\right)\right) = \frac{1}{2}\delta\left(x - \frac{1}{2}\right) = \frac{1}{2}\delta_{\frac{1}{2}}(x)$$

(d)

$$\delta'(ax) = \frac{1}{a|a|}\delta'(x)$$

↓

$$\delta'(2x) = \frac{1}{4}\delta'(x)$$

□

D.18

(a) $\delta * e^{-x^2}$

(b) $\delta' * (e^{-x}\theta(x))$

(c) $\delta_1 * \sin(2x)$

Solution:

Using the definition of the convolution

$$\varphi * U = \int \varphi(x-y)U(y) \, \mathrm{d}$$

(a)

$$\int e^{-(x-y)^2} \delta(y) \, dy = e^{-x^2}$$

(b)

$$\begin{aligned} \int e^{-(x-y)} \theta(x-y) \delta'(y) \, dy &= [e^{y-x} \theta(x-y) \delta(y)]_{-\infty}^{\infty} - \int (e^{y-x} \theta(x-y))' \delta(y) \, dy = \\ &= (e^{-x} \theta(x))' = -e^{-x} \theta(x) + e^{-x} \delta(x) = \delta(x) - e^{-x} \theta(x) \end{aligned}$$

(c)

$$\int \sin(2(x-y)) \delta(y-1) \, dy = \sin(2x-2)$$

□

D.22

Solution:

Definition of the cosine series expansion for distributions

$$\begin{aligned} U &= \frac{c_0}{2} + \sum_{k=1}^{\infty} c_k \cos\left(\frac{k\pi x}{L}\right), \quad 0 < x < L \\ c_k &= \frac{2}{L} \int_0^L U(x) \cos\left(\frac{k\pi x}{L}\right) \, dx \end{aligned}$$

We will also use the expansion of $f(x)\delta'(x) = f(0)\delta'(x) - f'(0)\delta(x)$.

This gives ($k \neq 0$)

$$\begin{aligned} c_k &= \frac{2}{\pi} \int_0^{\pi} \delta'\left(x - \frac{\pi}{2}\right) \cos(kx) \, dx = \frac{2}{\pi} \int_0^{\pi} \cos\left(k\frac{\pi}{2}\right) \delta'\left(x - \frac{\pi}{2}\right) + k \sin\left(k\frac{\pi}{2}\right) \delta\left(x - \frac{\pi}{2}\right) \, dx = \\ &= \frac{2}{\pi} \int_0^{\pi} k \sin\left(\frac{k\pi}{2}\right) \, dx = \frac{2}{\pi} k \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

and for $k = 0$:

$$c_0 = \frac{2}{\pi} \int_0^{\pi} \delta' \left(x - \frac{\pi}{2} \right) dx = \frac{2}{\pi} \left[\delta \left(x - \frac{\pi}{2} \right) \right]_0^{\pi} = 0$$

$$\delta' \left(x - \frac{\pi}{2} \right) = \frac{2}{\pi} \sum_{k=1}^{\infty} k \sin \left(\frac{k\pi}{2} \right) \cos(kx)$$

□

D.23

Solution:

Whut □

D.24

Solution:

Diffusion,

$$\begin{aligned} \frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < L, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(L, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= M \delta_{\frac{L}{2}}, & 0 < x < L \end{aligned}$$

We have homogenous Neumann boundary conditions, which should give us a solution in the form of a cosine series expansion.

$$u(x, t) = \sum u_k(t) \cos \left(\frac{k\pi x}{L} \right)$$

Differentiating termwise and putting into the differential equation gives

$$\sum (u'_k(t) + \alpha u_k(t)) \cos\left(\frac{k\pi x}{L}\right) = 0, \quad \alpha = \frac{Dk^2\pi^2}{L^2}$$

$$\downarrow$$

$$u'_k(t) + \alpha u_k(t) = 0$$

$$\downarrow$$

$$u_k(t) = c_k e^{-\alpha t}$$

We can also expand $u(x, 0) = M\delta_{\frac{L}{2}}$ in a cosine series, to get the coefficients, $u_k(0) = c_k$

$$u(x, 0) = M\delta_{\frac{L}{2}} = \sum a_k \cos\left(\frac{k\pi x}{L}\right)$$

$$c_0 = \frac{1}{L} \int_0^L M\delta_{\frac{L}{2}} dx = \frac{M}{L}$$

$$c_k = \frac{2}{L} \int_0^L M\delta_{\frac{L}{2}} \cos\left(\frac{k\pi x}{L}\right) dx = \frac{2M}{L} \cos\left(\frac{k\pi}{2}\right) = , \quad k = 1, 2, 3, \dots$$

This gives the full solution

$$u(x, t) = \frac{M}{L} \left(1 + 2 \sum_{k=1}^{\infty} \cos\left(\frac{k\pi}{2}\right) e^{-\alpha t} \cos\left(\frac{k\pi x}{L}\right) \right)$$

□

D.25

Solution:

□

D.27

Solve the three-dimensional potential problem, with $r = \sqrt{x^2 + y^2 + z^2}$

$$\Delta u = \delta(r - 2), \quad 1 < r < 3$$

$$u = 1, \quad r = 1$$

$$u = 3, \quad r = 3$$

Solution:

Symmetric in r so the Laplacian becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \delta(r-2)$$

$$\left(r^2 \frac{\partial u}{\partial r} \right)' = r^2 \delta(r-2) = 4\delta(r-2)$$

↓

$$r^2 \frac{\partial u}{\partial r} = 4\theta(r-2) + A$$

$$\frac{\partial u}{\partial r} = \frac{4}{r^2} \theta(r-2) + \frac{A}{r^2}$$

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{A}{r} + B$$

$$u(1) = 1 = -\theta(-1) - A + B = 0 \rightarrow -A + B = 1$$

$$u(3) = 3 = \left(\frac{2}{3} \right) \theta(1) - \frac{A}{3} + B = \frac{2}{3} + \frac{2}{3}A = 3$$

↓

$$\begin{cases} -A + B = 1 \\ \frac{2}{3} - \frac{A}{3} + B = 3 \end{cases} \rightarrow \frac{2}{3} + \frac{2}{3}A = 2 \rightarrow A = 2, B = 3$$

↓

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{2}{r} + 3$$

□

D.29**Solution:**

$$-\Delta u = -u + \delta(x-a)\delta(y-b), \quad 0 < x < 2, 0 < y < 1$$

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -u + \delta_{a,b}$$

□

D.30

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

(a) The Fourier transform of $\delta(x)$ is $\mathcal{F}[\delta(x)] = 1$ and the translation rule is $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)]$

$$\mathcal{F}[\delta(t-1)] = e^{-i\omega}$$

(b) The derivative rule is $\mathcal{F}[f'(t)] = i\omega \mathcal{F}[f(t)]$. Using this and the translation rule

$$\mathcal{F}[\delta'(t+2)] = i\omega e^{2i\omega}$$

(c) Fourier transform of a constant is $2\pi\delta(\omega)$

$$\mathcal{F}[2] = 4\pi\delta(\omega)$$

(d)

$$\sin(2t) = \frac{e^{2it} - e^{-2it}}{2i}$$

We have $\mathcal{F}[e^{i\omega_0 t} f(t)] = \mathcal{F}[f(\omega - \omega_0)]$

$$\mathcal{F}\left[\frac{e^{2it}}{2i} \cdot 1 - \frac{e^{-2it}}{2i} \cdot 1\right] = \frac{\pi}{i}(\delta(\omega - 2) - \delta(\omega + 2))$$

(e) $\mathcal{F}[tf(t)] = i \frac{d}{d\omega} \mathcal{F}[f(t)]$

$$\begin{aligned} \mathcal{F}[(t+1)^2] &= \mathcal{F}[t^2 + 2t + 1] = i^2 \frac{d^2}{d\omega^2} 2\pi\delta(\omega) + i \frac{d}{d\omega} 4\pi\delta(\omega) + 2\pi\delta(\omega) = \\ &= -2\pi\delta''(\omega) + 4\pi i\delta'(\omega) + 2\pi\delta(\omega) \end{aligned}$$

(f) This distribution is not tempered, and thus cannot be Fourier transformed. $\square$

D.31

- (a) $\delta(\omega-1)$
- (b) $\delta'(\omega+2)$

(c) 2

(d) $\sin(2\omega)$ (e) $(\omega + 1)^2$ **Solution:**

Going backwards on the identities gives

(a)

$$\begin{aligned}
\mathcal{F}[1] &= 2\pi\delta(\omega) \\
\mathcal{F}[e^{i\omega_0 t} \cdot 1] &= 2\pi\delta(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta(\omega - 1)] &= \frac{e^{it}}{2\pi}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathcal{F}[t] &= 2\pi i\delta'(\omega) \\
\mathcal{F}[e^{i\omega_0 t} t] &= 2\pi i\delta'(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta'(\omega + 2)] &= \frac{e^{-2it}}{2\pi i} t
\end{aligned}$$

(c)

$$\begin{aligned}
\mathcal{F}[\delta(t)] &= 1 \\
&\downarrow \\
\mathcal{F}^{-1}[2] &= 2\delta(t)
\end{aligned}$$

(d)

$$\begin{aligned}
\mathcal{F}[\theta(t+1) - \theta(t-1)] &= 2\frac{\sin(\omega)}{\omega} \\
\mathcal{F}\left[\theta\left(\frac{t}{2} + 1\right) - \theta\left(\frac{t}{2} - 1\right)\right] &= \mathcal{F}[\theta(t+2) - \theta(t-2)] = 2\frac{\sin(2\omega)}{\omega} \\
\mathcal{F}[(\theta(t+2) - \theta(t-2))'] &= \mathcal{F}[\delta(t+2) - \delta(t-2)] = i\omega\left(2\frac{\sin(2\omega)}{\omega}\right) = 2i\sin(2\omega) \\
&\downarrow \\
\mathcal{F}^{-1}[\sin(2\omega)] &= \frac{1}{2i}(\delta(t+2) - \delta(t-2))
\end{aligned}$$

(e)

$$\mathcal{F} [\delta'(t)] = i\omega$$

$$\mathcal{F} [-\delta''(t)] = \omega^2$$

$$\mathcal{F} [\delta(t)] = 1$$

$$\downarrow$$

$$\mathcal{F}^{-1} [\omega^2 + 2\omega + 1] = -\delta''(t) - 2i\delta'(t) + \delta(t)$$

□

D.34

Solution:

□

D.36

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

$$\mathcal{L} [\delta(t)] = 1$$

$$\mathcal{L} [f(t-t_0)] = e^{-t_0s} \mathcal{L} [f(t)]$$

$$\mathcal{L} [f'(t)] = s\mathcal{L} [f(t)]$$

$$\mathcal{L} [e^{at}f(t)] = \mathcal{L} [f(t-a)]$$

$$\mathcal{L} [\theta(t)] = \frac{1}{s}$$

(a)

$$\mathcal{L} [\delta(t-1)] = e^{-s}$$

(b)

$$\mathcal{L} [\delta'(t+1)] = se^{2s}$$

(c) Not possible

(d) Not possible

(e) Not possible

$$\mathcal{L} [(t+1)^2] = \mathcal{L} [t^2 + 2t + 1]$$

(f)

$$\mathcal{L} [e^t \theta(t)] = \frac{1}{s-1}, \quad \operatorname{Re} s > 0$$

□

D.41

Expand the differential equations to $\mathbb{R}$ so that the boundary conditions are fulfilled.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases}$$

Solution:

Dirichlet conditions give odd expansion, Neumann conditions give even expansions.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases} \rightarrow (u^-)'' - iu^- = 2\delta', \quad x \in \mathbb{R}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases} \rightarrow (u^+)'' - iu^+ = 2\delta, \quad x \in \mathbb{R}$$

□

Hilbert Spaces

H.1

Solution:

□

H.2

Solution:

□

H.3

Let $u = (2, 1 + i, i, 1 - i)$ and $v = (1, i, -1, i)$. Calculate

$$\|u\|, \|v\|, (u \mid v), (v \mid u)$$

with respect to the ordinary scalar product in $\mathbb{C}^4$.

Solution:

$$\|u\| = \sqrt{(u | u)} = \sqrt{(2, 1+i, i, 1-i) \cdot (2, 1+i, i, 1-i)} = \sqrt{4+2+1+2} = 3$$

$$\|v\| = \sqrt{(v | v)} = \sqrt{(1, i, -1, i) \cdot (1, i, -1, i)} = \sqrt{1+1+1+1} = 2$$

$$(u | v) = \overline{(2, 1+i, i, 1-i)} \cdot (1, i, -1, i) = 2+i+1+i+i-1 = 2+3i$$

$$(v | u) = \overline{(1, i, -1, i)} \cdot (2, 1+i, i, 1-i) = 2-i+1-i-i-1 = 2-3i$$

□

H.4

Let

$$f(x) = 1 + ix, \quad g(x) = 2 + ix^2, \quad 0 \leq x \leq 1.$$

Calculate $(f | g)$ and $(g | f)$, where the scalar product is taken in $L_2[0, 1]$.

Solution:

The scalar product is defined as

$$(u | v) = \int_L \bar{u}v \, dx$$

This gives

$$\begin{aligned} (f | g) &= \int_0^1 \overline{(1+ix)}(2+ix^2) \, dx = \int_0^1 (2+ix^2-2ix+x^3) \, dx = \\ &= \left[2x + \frac{ix^3}{3} - ix^2 + \frac{x^4}{4} \right]_0^1 = \frac{9}{4} - \frac{2}{3}i \end{aligned}$$

and

$$\begin{aligned} (g | f) &= \int_0^1 \overline{(2+ix^2)}(1+ix) \, dx = \int_0^1 (2+2ix-ix^2+x^3) \, dx = \\ &= \left[2x + ix^2 - \frac{ix^3}{3} + \frac{x^4}{4} \right]_0^1 = \frac{9}{4} + \frac{2}{3}i \end{aligned}$$

As expected,

$$(f | g) = \overline{(g | f)}.$$

□

H.5a

Motivate why the expression

$$\langle f | g \rangle = \int_0^1 f^2(x)g^2(x) \, dx$$

is not a scalar product for real, continuous functions on the interval $[0, 1]$

Solution:

The scalar product needs to be linear, which this expression is not. For example

$$\langle \lambda f | \mu g \rangle = \int_0^1 \lambda^2 f^2(x) \mu^2 g^2(x) \, dx$$

but linearity is

$$\lambda \mu \langle f | g \rangle = \lambda \mu \int_0^1 f^2(x)g^2(x) \, dx$$

□

H.11

Verify that the vectors

$$\begin{aligned} u_1 &= (1, 1, 1, 1) & u_2 &= (1, i, -1, -i) \\ u_3 &= (1, -1, 1, -1) & u_4 &= (1, -i, -1, i) \end{aligned}$$

are pairwise orthogonal in $\mathbb{C}^4$. Determine the corresponding orthonormal base e_1, e_2, e_3, e_4 and write

$$u = (2 + i, 1 + 2i, i, 1 - 4i)$$

in this base.

Solution:

The scalar product in $\mathbb{C}^4$ is

$$(u \mid v) = \overline{u} \cdot v$$

To check that the vectors are pairwise orthogonal, we check if their scalar product is 0:

$$(u_1 \mid u_2) = 1 + i - 1 - i = 0$$

$$(u_1 \mid u_3) = 1 - 1 + 1 - 1 = 0$$

$$(u_1 \mid u_4) = 1 - i - 1 + i = 0$$

$$(u_2 \mid u_3) = 1 + i - 1 - i = 0$$

$$(u_2 \mid u_4) = 1 - 1 + 1 - 1 = 0$$

$$(u_3 \mid u_4) = 1 + i - 1 - i = 0$$

This confirms they are pairwise orthogonal. All vectors have the length 2, so the orthonormal base is

$$e_1 = \frac{u_1}{2} \quad e_2 = \frac{u_2}{2}$$

$$e_3 = \frac{u_3}{2} \quad e_4 = \frac{u_4}{2}$$

To find the coordinates of u in terms of e_1, e_2, e_3 and e_4 , we use the projection formula. Since

$$u = \sum_{k=1}^4 c_k e_k$$

$$\downarrow$$

$$c_k = \frac{(e_k \mid u)}{(e_k \mid e_k)} = (e_k \mid u)$$

$$c_1 = (e_1 \mid u) = \frac{1}{2} \overline{(1, 1, 1, 1)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 2$$

$$c_2 = (e_2 \mid u) = \frac{1}{2} \overline{(1, i, -1, -i)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 4$$

$$c_3 = (e_3 \mid u) = \frac{1}{2} \overline{(1, -1, 1, -1)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 2i$$

$$c_4 = (e_4 \mid u) = \frac{1}{2} \overline{(1, -i, -1, i)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = -2$$

$$\downarrow$$

$$u = 2e_1 + 4e_2 + 2ie_3 - 2e_4$$

□

H.12

Let

$$f_n(x) = x^n.$$

Is $\{f_n\}_0^\infty$ pairwise orthogonal in

(a) $L_2(0, 1)$

(b) $L_2(-1, 1)$

Solution:

(a) No, since the scalar product in $L_2(0, 1)$ is not zero:

$$(x^j | x^k) = \int_0^1 x^{j+k} dx = \left[\frac{x^{j+k+1}}{j+k} \right]_0^1 = \frac{1}{j+k} \neq 0$$

(b) No,

$$(x^j | x^k) = \int_{-1}^1 x^{j+k} dx = \left[\frac{x^{j+k+1}}{j+k} \right]_{-1}^1 \rightarrow \begin{cases} = 0, & \text{if } j+k+1 \text{ even} \\ \neq 0, & \text{if } j+k+1 \text{ odd} \end{cases}$$

□

H.13

Let u, v in $L_2(0, 1)$ be given by

$$u(x) = 1, \quad 0 \leq x \leq 1,$$

$$v(x) = x, \quad 0 \leq x \leq 1.$$

Calculate the projection of v onto u , and the projection of u onto v .

Solution:

Using the projection formula, the projection of v onto u is

$$P_{[u]}v = \frac{(u | v)}{(u | u)}u = \frac{\int_0^1 x dx}{\int_0^1 dx} 1 = \frac{1}{2}$$

and the projection of u onto v is

$$P_{[v]}u = \frac{(v | u)}{(v | v)}v = \frac{\int_0^1 x \, dx}{\int_0^1 x^2 \, dx}x = \frac{3}{2}x$$

□

H.15

$$(u | v) = \int_{-\infty}^{\infty} u(x)v(x) \frac{e^{-x^2}}{\sqrt{\pi}} \, dx$$

Determine the first three Hermite polynomials

Solution:

Using Gram-Schmidt with $u_n = [1, x, x^2]$

$$\begin{aligned}\varphi_0 &= 1 \\ (\varphi_0 | \varphi_0) &= \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\sqrt{\pi}} \, dx = 1 \\ \psi_0 &= \frac{\varphi_0}{(\varphi_0 | \varphi_0)^{\frac{1}{2}}} = 1\end{aligned}$$

Next

$$\begin{aligned}\varphi_1 &= u_1 - P_{[\varphi_0]}u_1 = u_1 - \frac{(\varphi_0 | u_1)}{(\varphi_0 | \varphi_0)}\varphi_0 = x - \frac{(1 | x)}{(1 | 1)}1 = \\ &\quad \underbrace{x - \int_{-\infty}^{\infty} x \frac{e^{-x^2}}{\sqrt{\pi}} \, dx}_{=0 \text{ (odd function)}} = x \\ \psi_1 &= \frac{\varphi_1}{(\varphi_1 | \varphi_1)^{\frac{1}{2}}} = \frac{x}{(x | x)^{\frac{1}{2}}} \\ (x | x) &= \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} \, dx = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x)(-2xe^{-x^2}) \, dx = \\ &= -\frac{1}{2\sqrt{\pi}} \left([xe^{-x^2}]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-x^2} \, dx \right) = \frac{1}{2}\end{aligned}$$

Finally this gives

$$\psi_1 = \frac{x}{(x | x)^{\frac{1}{2}}} = \sqrt{2}x$$

Next

$$\begin{aligned}
 \varphi_2 &= u_2 - P_{[\varphi_0, \varphi_1]} u_2 = x^2 - \frac{(1 \mid x^2)}{(1 \mid 1)} 1 - \frac{(x \mid x^2)}{(x \mid x)} x = \\
 &= x^2 - \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx - \underbrace{\int_{-\infty}^{\infty} x^3 \frac{e^{-x^2}}{\sqrt{\pi}} dx}_{=0 \text{ odd function}} = x^2 - \frac{1}{2} \\
 \psi_2 &= \frac{\varphi_2}{(\varphi_2 \mid \varphi_2)^{\frac{1}{2}}} = \left(x^2 - \frac{1}{2}\right) \left(\int_{-\infty}^{\infty} \left(x^2 - \frac{1}{2}\right)^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx \right)^{-\frac{1}{2}} = \\
 &= \left(x^2 - \frac{1}{2}\right) \left(\int_{-\infty}^{\infty} \left(x^4 - x^2 + \frac{1}{4}\right) \frac{e^{-x^2}}{\sqrt{\pi}} dx \right)^{-\frac{1}{2}} \\
 &\quad \text{(i) } \int_{-\infty}^{\infty} \frac{1}{4} \frac{e^{-x^2}}{\sqrt{\pi}} dx = \boxed{\frac{1}{4}} \\
 &\quad \text{(ii) } \int_{-\infty}^{\infty} -x^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx = \boxed{-\frac{1}{2}} \\
 &\quad \text{(iii) } \int_{-\infty}^{\infty} x^4 \frac{e^{-x^2}}{\sqrt{\pi}} dx = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x^3)(-2xe^{-x^2}) dx = \\
 &\quad = \underbrace{\left[x^2 e^{-x^2}\right]_{-\infty}^{\infty}}_{=0} - \underbrace{\int_{-\infty}^{\infty} 3x^2 e^{-x^2} dx}_{=\frac{3\sqrt{\pi}}{2}} = \boxed{\frac{3}{4}}
 \end{aligned}$$

So

$$\psi_2 = \left(x^2 - \frac{1}{2}\right) \left(\frac{1}{4} - \frac{1}{2} + \frac{3}{4}\right)^{-\frac{1}{2}} = \left(x^2 - \frac{1}{2}\right) \sqrt{2}$$

The three first orthonormal Hermite polynomials are

$$\begin{aligned}
 \psi_0 &= \boxed{1} \\
 \psi_1 &= \boxed{\sqrt{2}x} \\
 \psi_2 &= \boxed{\sqrt{2}\left(x^2 - \frac{1}{2}\right)}
 \end{aligned}$$

□

H.18

$$\int_0^\pi (x - a \sin(x) - b \sin(2x))^2 dx$$

is minimized.

Solution:

$$v \in \mathcal{M} = [\sin(x), \sin(2x)]$$

$\sin(x)$ and $\sin(2x)$ are orthogonal and we can use the projection formula which says that the minimum is obtained when

$$v = P_{[\mathcal{M}]} u$$

$$v_1 = a \sin(x) = P_{[\sin(x)]} u = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))} \sin(x)$$

$$v_2 = b \sin(2x)$$

From this, a is calculated as

$$a = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))}$$

$$(\sin(x) | x) = \int_0^\pi x \sin(x) dx = -[x \cos(x)]_0^\pi + \int_0^\pi \pi \cos(x) dx = \pi$$

$$(\sin(x) | \sin(x)) = \int_0^\pi \sin^2 x dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$a = \frac{\pi}{\frac{\pi}{2}} = 2$$

and b is calculated in the same way as

$$b = \frac{(\sin(2x) | x)}{(\sin(2x) | \sin(2x))}$$

$$(\sin(2x) | x) = \int_0^\pi x \sin(2x) dx = -\left[\frac{x \cos(2x)}{2}\right]_0^\pi + \int_0^\pi \frac{\cos(2x)}{2} dx = -\frac{\pi}{2}$$

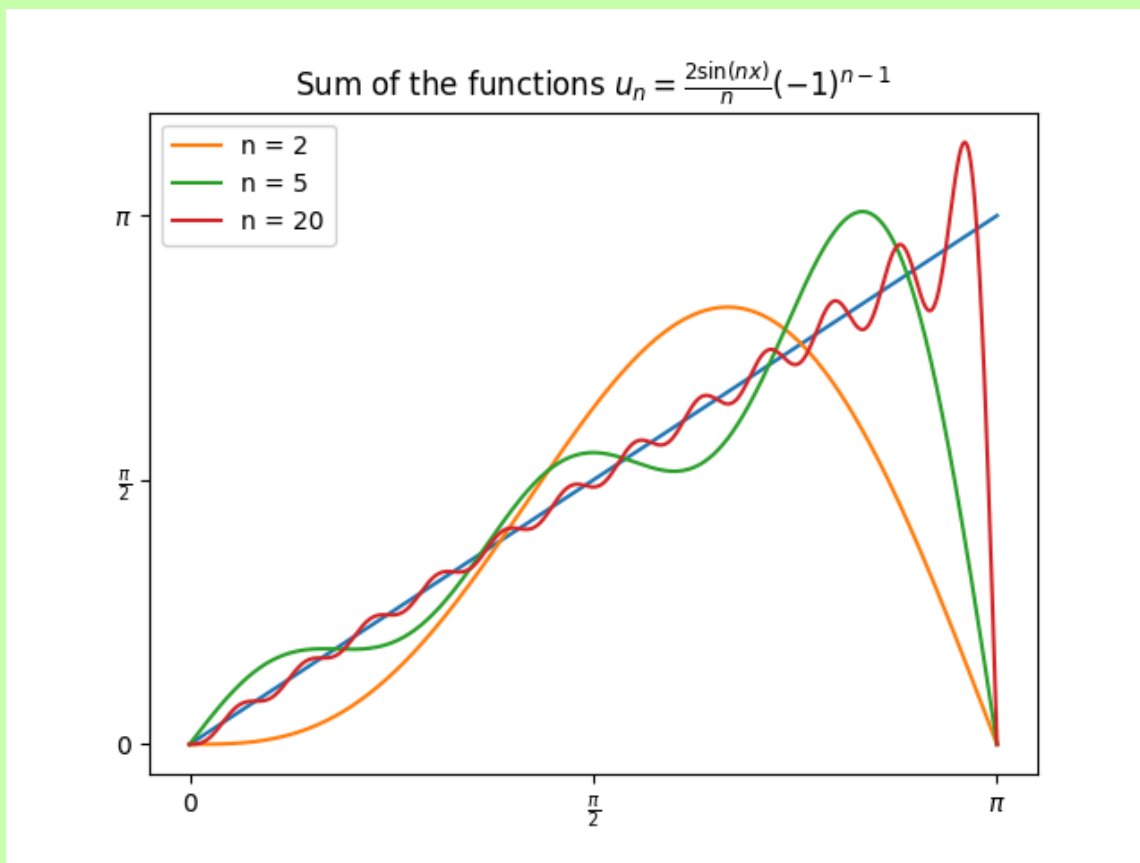
$$(\sin(2x) | \sin(2x)) = \int_0^\pi \sin^2(2x) dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$b = -\frac{\frac{\pi}{2}}{\frac{\pi}{2}} = -1$$

In general, the functions $v \in \mathcal{M} = \{\sin(nx)\}_{n=1}^{\infty}$ that best approximate $u(x) = x$ at the interval $[0, \pi]$ is given by

$$u_n = \frac{2(-1)^{n-1} \sin nx}{n}$$



□

H.19

(a) Give an orthogonal base in $\mathcal{V}$ with the scalar product

$$(u | v) = \int_{-1}^1 u(x)v(x) \, dx$$

(b) Let $u(x) = x^2 - x$. Determine $v \in \mathcal{V}$ so that

$$\int_{-1}^1 |u(x) - v(x)|^2 \, dx$$

is minimized.

(c) Determine the lowest possible value of the integral in (b).

Solution:(a) $\square$ **H.20****Solution:**

For this setup, the scalar product is defined as

$$(u | v) = \int_0^1 u(x)v(x)w(x) \, dx = \int_0^1 u(x)v(x)x \, dx$$

Find an orthogonal base

$$\begin{aligned} \varphi_0 &= 1 \\ \varphi_1 &= x - P_{[\varphi_0]}x = x - \frac{(\varphi_0 | x)}{(\varphi_0 | \varphi_0)}\varphi_0 = x - 2 \int_0^1 x^2 \, dx = x - \frac{2}{3} \end{aligned}$$

Now $v \in \mathcal{M} = \{1, x - \frac{2}{3}\}$

Are looking for

$$\int_0^1 (x^4 - a\varphi_0 - b\varphi_1)^2 x \, dx$$

From which we can calculate

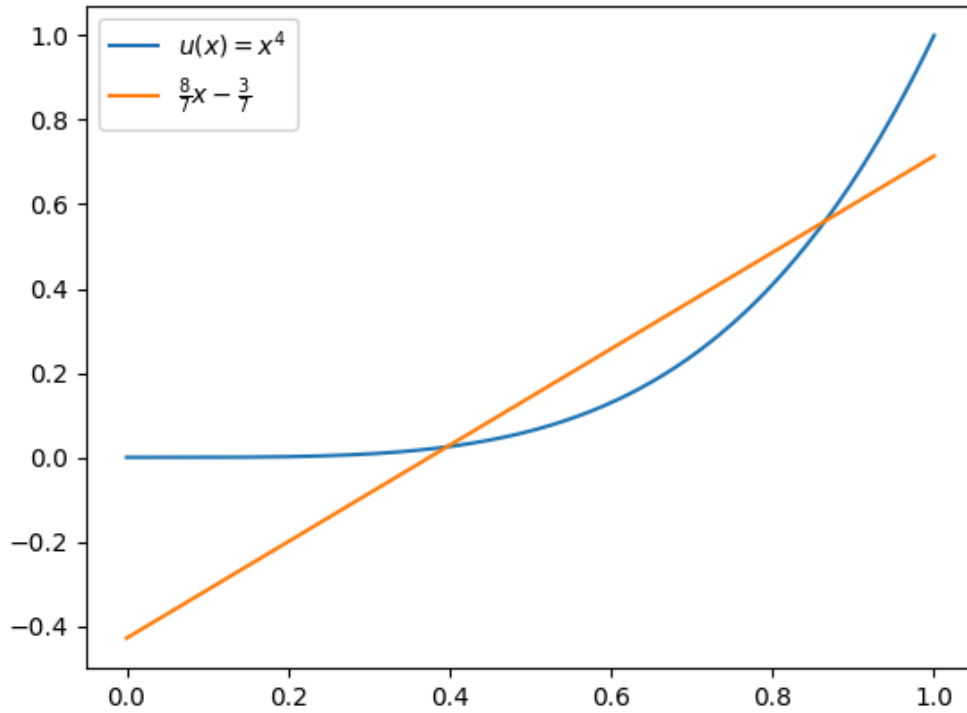
$$a = \frac{(\varphi_0 | x^4)}{(\varphi_0 | \varphi_0)} = \frac{\int_0^1 x^5 \, dx}{\int_0^1 x \, dx} = \frac{1}{3}$$

and

$$b = \frac{(\varphi_1 | x^4)}{(\varphi_1 | \varphi_1)} = \frac{\int_0^1 (x^6 - \frac{2}{3}x^5) \, dx}{\int_0^1 (x^3 - \frac{4}{3}x^2 + \frac{4}{9}x) \, dx} = \frac{8}{7}$$

The first degree polynomial that best approximates the function $u(x) = x^4$ in $I = [0, 1]$ is

$$\frac{1}{3}\varphi_0 + \frac{8}{7}\varphi_1 = \frac{8}{7}x - \frac{3}{7}$$



(b) We are now looking for a second degree polynomial which means our $\varphi \in \mathcal{M} = \{1, x, x^2\}$ φ_0 and φ_1 were already calculated in (a) so

$$\begin{aligned}\varphi_2 &= u_2 - P_{[\varphi_0, \varphi_1]} u_2 = x^2 - \frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 - \frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1 \\ \frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 &= 2(\varphi_0 | x^2) = 2 \int_0^1 x^3 dx = \frac{1}{2} \\ \frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1 &= (\varphi_1 | x^2)(36x - 24)\end{aligned}$$

Now calculating

$$(\varphi_1 | x^2) = \int_0^1 \left(x^4 - \frac{2x^3}{3} \right) dx = \frac{1}{30}$$

which gives finally

$$\varphi_2 = x^2 - \frac{6}{5}x + \frac{3}{10}$$

$$(\varphi_2 | \varphi_2) = \int_0^1 \left(x^2 - \frac{6x}{5} + \frac{3}{10} \right)^2 x \, dx = \frac{1}{600}$$

$$\varphi_2 = 600x^2 - 720x + 180$$

We can now find the best approximation with

$$\int_0^1 (x^4 - a\varphi_0 - b\varphi_1 - c\varphi_2)^2 x \, dx$$

$$c\varphi_2 = \frac{(\varphi_2 | x^4)}{(\varphi_2 | \varphi_2)} \varphi_2$$

$$c = \frac{(\varphi_2 | x^4)}{(\varphi_2 | \varphi_2)}$$

$$(\varphi_2 | x^4) = \int_0^1 (600x^2 - 720x + 180)x^4 \, dx = \frac{15}{7}$$

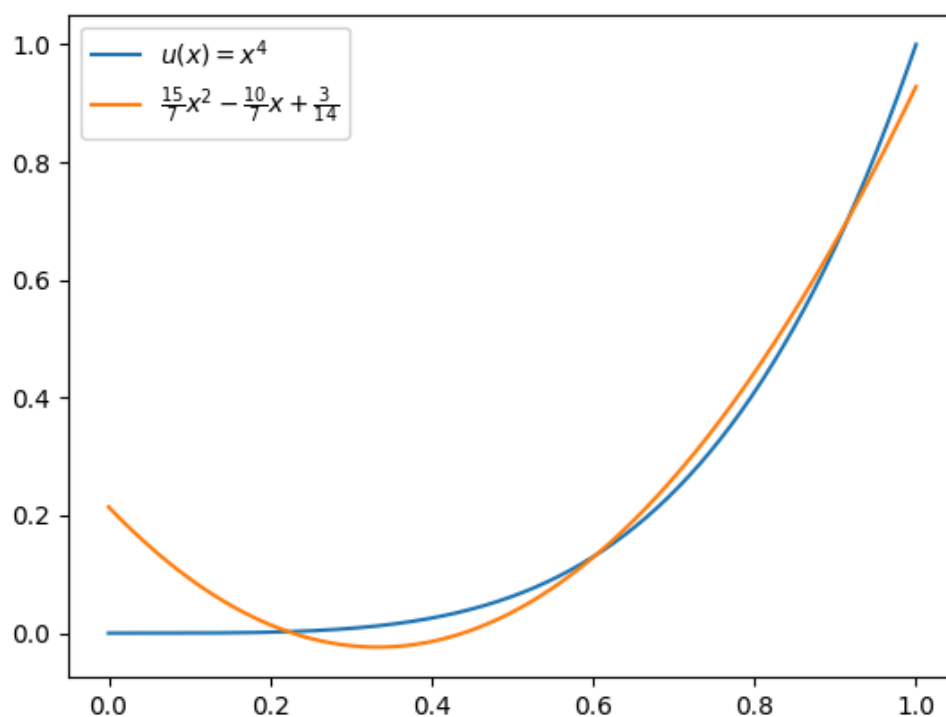
$$(\varphi_2 | \varphi_2) = \int_0^1 (600x^2 - 720x + 180)^2 x \, dx = 600$$

$$\downarrow$$

$$c = \frac{15}{4200} = \frac{1}{280}$$

The best second degree polynomial to approximate $u(x) = x^4$ is then

$$a\varphi_0 + b\varphi_1 + c\varphi_2 = \frac{15}{7}x^2 - \frac{10}{7}x + \frac{3}{14}$$



□

H.27

- (a) Show that p_n is orthogonal to all polynomials of degree $p < n$
- (b) Show that if p is a polynomial, not identically zero, such that $(p_k | p) = 0$ for $k = 0, 1, 2$, then degree of $p \geq 3$.

Solution:

- (a) All polynomials of degree m can be written

$$p = \sum_{k=0}^m c_k p_k$$

Gives

$$(p_n | p) = \left(p_n | \sum_{k=0}^m c_k p_k \right) = \sum_{k=0}^m c_k (p_n | p_k) = 0 \text{ for } m < n$$

since all polynomials are pairwise orthogonal

(b)

$$p = \sum_{k=0}^m c_k p_k$$

where

$$c_k = \frac{(p_k | p)}{(p_k | p_k)}$$

$$(p_k | p) = 0, \text{ for } k = 0, 1, 2 \rightarrow c_0 = c_1 = c_2 = 0$$

The restriction that $p \neq 0$ gives that $\deg p \geq 3$ $\square$

H.30

Solution:

$\square$

H.31

Solution:

$\square$

H.33

$$f_n(x) = x^n, \quad x \in I = [0, 1], \quad n = 0, 1, \dots$$

Determine $\|f_n\|_p$ for $p = 1, 2$ and ∞ . Is it for any of these norms true that $f_n \rightarrow 0$ when $n \rightarrow \infty$? What is the pointwise limit function?

Solution:

$$\|f_n\|_1 = \int_0^1 |x^n| dx = \left[\frac{x^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1} \rightarrow 0, \quad n \rightarrow \infty$$

$$\|f_n\|_2 = \left(\int_0^1 |x^n|^2 dx \right)^{\frac{1}{2}} = \left(\left[\frac{x^{2n+1}}{2n+1} \right]_0^1 \right)^{\frac{1}{2}} = \frac{1}{\sqrt{2n+1}} \rightarrow 0, \quad n \rightarrow \infty$$

$$\|f_n\|_\infty = \sup_{x \in [0,1]} |f_n| = \sup_{x \in [0,1]} |x^n| = 1 \rightarrow 1, \quad n \rightarrow \infty$$

□

H.36**Solution:**

□

H.38**Solution:**

□

H.39

A linear operator $\mathcal{A}$ on $\mathcal{H}$ is said to be unitary if it is bijective and

$$(\mathcal{A}u \mid \mathcal{A}v) = (u \mid v)$$

for all $u, v \in D_{\mathcal{A}}$. Show that every eigenvalue to a unitary operator has magnitude one.

Solution:

$$\begin{aligned}
\mathcal{A}u &= \lambda u, \quad \mathcal{A}v = \lambda v \\
&\downarrow \\
(\mathcal{A}u \mid \mathcal{A}v) &= (\lambda u \mid \lambda v) = \bar{\lambda}\lambda(u \mid v) = |\lambda|^2 (u \mid v) = (u \mid v) \\
&\downarrow \\
|\lambda|^2 &= 1
\end{aligned}$$

□

H.45**Solution:**

□

H.47

Show with the help of the definition that

$$\mathcal{A} = -\frac{d^2}{dx^2}, \quad D_{\mathcal{A}} = \{u \in \mathcal{C}^2[0, 1] \mid u'(0) = u(1) = 0\}$$

is a symmetric positive semidefinite operator on $L_2[0, 1]$. Determine its eigenvalues and eigenvectors.

Solution:

Symmetric:

$$\begin{aligned}
(u \mid \mathcal{A}v) &= - \int_0^1 \bar{u} v'' \, dx = - \underbrace{[\bar{u} v']_0^1}_{=0} + \int_0^1 \bar{u}' v' \, dx = \\
(*) &= \int_0^1 \bar{u}' v' \, dx = \underbrace{[\bar{u}' v]_0^1}_{=0} - \int_0^1 \bar{u}'' v \, dx = (\mathcal{A}u \mid v)
\end{aligned}$$

Positive semidefinite, start from (*) and set $u = v$:

$$(*) = \int_0^1 \bar{u}' u' \, dx = \int_0^1 |u'|^2 \, dx \geq 0$$

Eigenvalues and eigenvectors:

$$\mathcal{A}u = \lambda u \rightarrow -u'' - \lambda u = 0$$

$$\downarrow$$

$$u'' + \lambda u \rightarrow u(x) = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$$

$$u'(0) = -\sqrt{\lambda}A \sin(\sqrt{\lambda} \cdot 0) + \sqrt{\lambda}B \cos(\sqrt{\lambda} \cdot 0) = 0 \rightarrow B = 0$$

$$u(1) = A \cos(\sqrt{\lambda} \cdot 1) = 0 \rightarrow \sqrt{\lambda} = \frac{\pi}{2} + k\pi$$

$$\downarrow$$

$$\lambda_k = \frac{\pi^2}{4} + k^2\pi^2 + k\pi^2$$

$$\varphi_k(x) = \cos(\sqrt{\lambda}x)$$

□

H.50

$$\mathcal{A} = -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \right)$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(-1, 1) \mid u \text{ and } u' \text{ bounded/limited}\}$$

- Determine a scalar product in which $\mathcal{A}$ is symmetric.
- Show that $\mathcal{A}$ is symmetric in the scalar product from (a), using the definition.
- Show that $1, x, 2x^2 - 1$ are eigenfunctions to $\mathcal{A}$ and determine the corresponding eigenvalues.
- Evaluate

$$\int_{-1}^1 \frac{2x^2 - 1}{\sqrt{1-x^2}} dx$$

Solution:

- The operator is a Sturm-Liouville operator

$$\mathcal{A}u = -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \right) = -\frac{1}{w} (pu')', \quad w(x) = \frac{1}{\sqrt{1-x^2}}$$

and the scalar product is then

$$(u \mid v) = \int_{-1}^1 \overline{u(x)} v(x) \frac{1}{w(x)} dx$$

- Using the scalar product definition

$$(u \mid v) = \int \overline{u(x)} v(x) w(x) \, dx, \quad w(x) = \frac{1}{\sqrt{1-x^2}}$$

we get

$$\begin{aligned}
 (u \mid \mathcal{A}v) &= \\
 &= \int_{-1}^1 \overline{u(x)} [\mathcal{A}v(x)] \frac{1}{w(x)} \, dx = \int_{-1}^1 \overline{u(x)} \left[-\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} v(x) \right) \right] \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= - \int_{-1}^1 \overline{u(x)} \left(\sqrt{1-x^2} v'(x) \right)' \, dx = - \underbrace{\left[\overline{u(x)} \sqrt{1-x^2} v'(x) \right]_{-1}^1}_{=0} + \int_{-1}^1 \overline{u'(x)} \sqrt{1-x^2} v'(x) \, dx = \\
 &= \int_{-1}^1 \left(\overline{u'(x)} \sqrt{1-x^2} \right) v'(x) \, dx = \underbrace{\left[\left(\overline{u'(x)} \sqrt{1-x^2} \right) v(x) \right]_{-1}^1}_{=0} - \int_{-1}^1 \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, dx = \\
 &= \int_{-1}^1 - \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, dx = \int_{-1}^1 -\sqrt{1-x^2} \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= \int_{-1}^1 \left[-\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \overline{u(x)} \right) \right] v(x) \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= \int_{-1}^1 [\mathcal{A}u(x)] v(x) \frac{1}{w(x)} \, dx = (\mathcal{A}u \mid v)
 \end{aligned}$$

(c)

$$\mathcal{A}u = \lambda u$$

$$u(x) = 1$$

$$\downarrow$$

$$\mathcal{A} \cdot 1 = \lambda_0 \cdot 1 \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \cdot 1 \right) = \lambda_0 \cdot 1 \rightarrow 0 = \lambda_0 \cdot 1 \rightarrow \boxed{\lambda_0 = 0}$$

$$u(x) = x$$

$$\downarrow$$

$$\mathcal{A}x = \lambda_1 x \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} x \right) = \lambda_1 x \rightarrow -\sqrt{1-x^2} \frac{-2x}{2\sqrt{1-x^2}} = \lambda_1 x \rightarrow$$

$$\rightarrow x = \lambda_1 x \rightarrow \boxed{\lambda_1 = 1}$$

$$u(x) = 2x^2 - 1$$

$$\downarrow$$

$$\begin{aligned} \mathcal{A}(2x^2 - 1) &= \lambda_2(2x^2 - 1) \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} (2x^2 - 1) \right) = \lambda_2(2x^2 - 1) \rightarrow \\ &\rightarrow -\sqrt{1-x^2} \frac{d}{dx} (4x\sqrt{1-x^2}) = -\sqrt{1-x^2} \left(4\sqrt{1-x^2} - \frac{8x^2}{2\sqrt{1-x^2}} \right) = -4 + 4x^2 + 4x^2 = \\ &= 8x^2 - 4 = \lambda_2(2x^2 - 1) \rightarrow 4(2x^2 - 1) = \lambda_2(2x^2 - 1) \rightarrow \boxed{\lambda_2 = 4} \end{aligned}$$

(d) The integral can be rewritten with the scalar product in (a) as

$$\int_{-1}^1 \frac{2x^2 - 1}{\sqrt{1-x^2}} dx = (1 \mid 2x^2 - 1), \quad w(x) = \frac{q}{\sqrt{1-x^2}}$$

Since the functions 1 and $2x^2 - 1$ are eigenfunctions, the scalar product between them are pairwise orthogonal and therefore = 0.

□

H.51

Solution:

□

H.52**Solution:**

□

Operator

$$\mathcal{A} = -\text{laplace} = -\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$$

Solution:

$$\begin{aligned}
(u \mid \mathcal{A}v) &= \int_0^a \int_0^b \bar{u}(-v''_{xx} - v''_{yy}) \, dx \, dy = \\
(1) &= - \int_0^b \underbrace{\left(\int_0^a \bar{u} v''_{xx} \, dx \right)}_{(i-x)} \, dy - \int_0^a \underbrace{\left(\int_0^b \bar{u} v''_{yy} \, dy \right)}_{(i-y)} \, dx \\
(i-x) &= \underbrace{[\bar{u} v'_x]_0^a}_{=0} - \int_0^a \bar{u}'_x v'_x \, dx \\
(i-y) &= \underbrace{[\bar{u} v'_y]_0^b}_{=0} - \int_0^b \bar{u}'_y v'_y \, dy \\
&\quad \downarrow \\
(1) &= \int_0^b \underbrace{\left(\int_0^a \bar{u}'_x v'_x \, dx \right)}_{(ii-x)} \, dy + \int_0^a \underbrace{\left(\int_0^b \bar{u}'_y v'_y \, dy \right)}_{(ii-y)} \, dx = (***) \\
(ii-x) &= \underbrace{[\bar{u}'_x v'_x]_0^a}_{=0} - \int_0^a \bar{u}''_{xx} v \, dx \\
(ii-y) &= \underbrace{[\bar{u}'_y v'_y]_0^b}_{=0} - \int_0^b \bar{u}''_{yy} v \, dy \\
&\quad \downarrow \\
(1) &= - \int_0^b \int_0^a \bar{u}''_{xx} v \, dx \, dy - \int_0^a \int_0^b \bar{u}''_{yy} v \, dy \, dx = \int_0^a \int_0^b (-\bar{u}''_{xx} - \bar{u}''_{yy}) v \, dx \, dy = \\
&= \boxed{(\mathcal{A}u \mid v)}
\end{aligned}$$

To show positive semidefinite, use the form described on (***) and $u = v$:

$$(***) = \int_0^a \int_0^b (\bar{u}'_x u'_x + \bar{u}'_y u'_y) \, dx \, dy = \int_0^a \int_0^b (|u'_x|^2 + |u'_y|^2) \, dx \, dy \geq 0$$

To find the eigenvalues associated with the operator, use separation of variables

$$\begin{aligned}
u(x, y) &= X(x)Y(y) \\
\mathcal{A}u &= \lambda u \rightarrow -X''Y - XY'' = \lambda XY \rightarrow -\frac{X''}{X} - \frac{Y''}{Y} = \lambda \\
&\quad \downarrow \\
&\begin{cases} \frac{X''}{X} = -\gamma_x^2, & X(0) = X(a) = 0 \\ \frac{Y''}{Y} = -\gamma_y^2, & Y'(0) = Y'(b) = 0 \end{cases}
\end{aligned}$$

Starting with $X(x)$:

$$X'' = -\gamma_x^2 X \rightarrow X(x) = e^{i\gamma_x x} = A \cos(\gamma_x x) + B \sin(\gamma_x x)$$

$$X(0) = A = 0 \rightarrow A = 0$$

$$X(a) = B \sin(\gamma_x a) = 0 \rightarrow \gamma_x = \frac{\pi k_x}{a}, k_x = 1, 2, 3 \dots$$

↓

$$X(x) = B \sin\left(\frac{\pi k_x x}{a}\right) \quad \square$$

Now $Y(y)$:

$$Y'' = -\gamma_y^2 Y \rightarrow Y(y) = e^{i\gamma_y y} = A \cos(\gamma_y y) + B \sin(\gamma_y y)$$

$$Y'(x) = -\gamma_y A \sin(\gamma_y y) + \gamma_y B \cos(\gamma_y y)$$

$$Y'(0) = \gamma_y B = 0 \rightarrow B = 0$$

$$Y'(b) = -\gamma_y A \sin(\gamma_y b) = 0 \rightarrow \gamma_y = \frac{\pi k_y}{b}, k_y = 0, 1, 2 \dots$$

↓

$$Y(y) = A \cos\left(\frac{\pi k_y y}{b}\right) \quad \square$$

□

H.8 Operators in Hilbert Spaces

Linear avbildning

$$\mathcal{A} : \mathcal{H}_1 \rightarrow \mathcal{H}_2$$

where $\mathcal{H}_1$ and $\mathcal{H}_2$ are two pre-Hilbert Spaces. Linearity:

$$\mathcal{A}(\lambda u + \mu v) = \lambda \mathcal{A}u + \mu \mathcal{A}v$$

is termed a **linear operator**.

Example H.19

Let I be a compact interval on $\mathbb{R}$, and let $K(x, y)$ be continuous for $x, y \in I$.

$$v(x) = \int_I K(x, y) u(y) \, dy$$

This defines an **integral operator**

$$\mathcal{K} : L_2(I) \rightarrow L_2(I)$$

The function $K(x, y)$ is termed the operators **kernel**

Special Functions

S.1

Solution:

Using the definition of $\Gamma(z)$:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

it can be shown that $\Gamma(z+1) = z\Gamma(z)$:

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt = \underbrace{[-t^z e^{-t}]_0^{\infty}}_{=0} + \int_0^{\infty} z t^{z-1} e^{-t} dt = \\ &= z \int_0^{\infty} t^{z-1} e^{-t} dt = z\Gamma(z) \end{aligned}$$

We also have

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \Gamma(3) = 2$$

Next

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \rightarrow [t = x^2 \rightarrow dt = 2x dx] = 2 \int_0^{\infty} x x^{-1} e^{-x^2} dx = \\ &= 2 \int_0^{\infty} e^{-x^2} dx = 2 \frac{\sqrt{\pi}}{2} \rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

Finally we will use the definition for $\Gamma(z)$, $\text{Re } z \leq 0$:

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)\dots(z+1)z}, \quad n \text{ chosen such that } \operatorname{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

□

S.5

Solve the Abelian integral

$$\int_0^x \frac{u(y)}{\sqrt{x-y}} dy = 1, \quad x > 0$$

Solution:

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = F(s)$$

↓

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = \frac{1}{s}$$

↓

$$U(s) = \frac{1}{\Gamma\left(\frac{1}{2}\right)s^{\frac{1}{2}}} = \frac{1}{\Gamma\left(\frac{1}{2}\right)^2} \frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} \xrightarrow{\mathcal{L}^{-1}} u(x)\theta(x) = \frac{1}{\pi} x^{-\frac{1}{2}}\theta(x)$$

↓

$$u(x) = \frac{1}{\pi\sqrt{x}}, \quad x > 0$$

□

S.6

Solution:

□

S.7

Show that

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

Solution:

Using the Fourier expansion

$$e^{ir \sin \theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |e^{ir \sin \theta}|^2 d\theta = \sum_{-\infty}^{\infty} |J_n(r)|^2$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{e^{ir \sin \theta}} e^{ir \sin \theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta = \frac{1}{2\pi} 2\pi = 1$$

↓

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

□

S.11

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate $J_{\frac{3}{2}}(x)$ (c) Show that $J_0'(x) = -J_1(x)$

Solution:

(a) The power series is

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k$$

$$\downarrow$$

$$\frac{J_\nu(x)}{x^\nu} = \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k$$

This can be differentiated termwise as

$$\begin{aligned} \frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{x}{2} \left(\frac{1}{2}\right)^\nu \sum_{\substack{k=1 \\ *}}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} \rightarrow [k \rightarrow k+1] \rightarrow \\ &\rightarrow -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + 1 + k + 1)} \left(-\frac{x^2}{4}\right)^k = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}} \end{aligned}$$

* (The $k=0$ contribution is zero, since it is a constant and was differentiated.)

(b) Setting $\nu = \frac{1}{2}$ we get

$$\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}}$$

$$\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\frac{\sqrt{\frac{2}{\pi x}} \sin x}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\sqrt{\frac{2}{\pi}} \frac{\sin x}{x} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{x \cos(x) - \sin(x)}{x^2} \right)$$

$$\downarrow$$

$$J_{\frac{3}{2}}(x) = -x^{\frac{1}{2}} \frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = x^{\frac{1}{2}} \sqrt{\frac{2}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^2} \right) = \sqrt{\frac{2x}{\pi}} \left(\frac{\sin x - x \cos x}{x^2} \right)$$

(c) Using the result from (a)

$$\frac{d}{dx} \left(\frac{J_0(x)}{x^0} \right) = -\frac{J_1(x)}{x^0} \rightarrow \boxed{J'_0(x) = -J_1(x)}$$

□

S.12**Solution:**

□

S.13

(a) $r^2 y'' + r y' = 0$

(b) $r^2 y'' + r y' - y = 0$

(c) $r^2 y'' + r y' + r^2 y = 0$

Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r}u' + \left(\lambda - \frac{\nu^2}{r^2}\right)u = 0$$

are

$$\begin{aligned} & aJ_\nu(\sqrt{\lambda}r) + bY_\nu(\sqrt{\lambda}r), & \text{if } \lambda > 0 \\ & ar^\nu + br^{-\nu}, & \text{if } \lambda = 0, \nu \neq 0 \\ & a + b \ln r, & \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2 y'' + r y' = 0 \rightarrow y'' + \frac{1}{r}y' = 0, \quad \lambda = \nu = 0$$

$$y(r) = a + b \ln r$$

(b)

$$r^2 y'' + r y' - y = 0 \rightarrow y'' + \frac{1}{r}y' - \frac{1}{r^2}y = 0, \quad \lambda = 0, \nu = 1$$

$$y(r) = ar + \frac{b}{r}$$

(c)

$$r^2 y'' + r y' + r^2 y = 0 \rightarrow y'' + \frac{1}{r} y' + y = 0, \quad \lambda = 1, \nu = 0$$

$$y(r) = a J_0(r) + b Y_0(r)$$

□

S.14

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

- (a) Determine a scalar product in which $\mathcal{A}$ is symmetric and positive semidefinite
- (b) Determine the eigenvalues and eigenvectors of $\mathcal{A}$.
- (c) Determine numerically the three lowest eigenvalues.
- (d) Let $u(x) = x$. Write down u 's expansion in eigenfunctions to $\mathcal{A}$. The coefficients should be given in integral form.

Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)} \mathcal{A}v(x) w(x) \, dx$$

with $w(x) = x$

$$\begin{aligned} (u \mid \mathcal{A}v) &= - \int_0^2 \overline{u} (xv')' \, dx = - \underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u'}x) v' \, dx = \\ &= \underbrace{\int_0^2 (x\overline{u'}) v' \, dx}_{= (*)} = \underbrace{[(x\overline{u'})v]_0^2}_{=0} - \int_0^2 (x\overline{u'})' v \, dx = (\mathcal{A}u \mid v) \end{aligned}$$

To show that it's positive semidefinite, start from (*) and set $u = v$:

$$\int_0^2 x \overline{u'} u' \, dx = \int_0^2 x |u'|^2 \, dx \geq 0$$

(b) Setting

$$\mathcal{A}u = \lambda u \rightarrow -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u$$

$$\downarrow$$

$$-\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u = 0 \rightarrow u'' + \frac{1}{x}u' + \lambda u = 0$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$aJ_0(\sqrt{\lambda}x) + bY_0(\sqrt{\lambda}x), \quad \lambda > 0, \nu = 0$$

Since Y_0 is unbounded near $x = 0$, b has to be zero to fulfill the requirement of u being bounded near zero.

$$u(x) = aJ_0(\sqrt{\lambda}x)$$

$$u'(2) = \sqrt{\lambda}aJ_0'(\sqrt{\lambda}2) = 0$$

This gives $\lambda_0 = 0$. The rest of the eigenvalues are given by

$$J_0'(2\sqrt{\lambda_k}) = 0 \rightarrow \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\lambda_0 = 0$$

$$\lambda_1 = \left(\frac{3.8317}{2}\right)^2 = 3.67$$

$$\lambda_2 = \left(\frac{7.0156}{2}\right)^2 = 12.31$$

□

S.15

Determine a non-trivial solution to the eigenvalue problem

$$\begin{aligned} -\Delta u &= \lambda u, \text{ in } \Omega \\ u &= 0, \quad \text{on } \partial\Omega \end{aligned}$$

where Ω is the circle sector $0 < r < 1, 0 < \theta < \frac{\pi}{3}$

Solution:

We have the conditions for $u(r, \theta)$

$$\begin{aligned} u(r, 0) &= u\left(r, \frac{\pi}{3}\right) = 0 \\ u(1, \theta) &= 0 \end{aligned}$$

The Laplacian in cylindrical coordinates is

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$$

Expressing $u(r, \theta)$ with separation of variables gives

$$u(r, \theta) = R(r)\Theta(\theta)$$

Putting this into the Laplacian gives

$$\begin{aligned} R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' &= -\lambda R\Theta \\ \downarrow \\ \frac{R''}{R} + \frac{1}{r}\frac{R'}{R} + \frac{1}{r^2}\frac{\Theta''}{\Theta} &= -\lambda \\ \downarrow \\ r^2\frac{R''}{R} + r\frac{R'}{R} + \lambda r^2 &= -\frac{\Theta''}{\Theta} \\ \downarrow \\ \frac{r^2}{R}\left(R'' + \frac{1}{r}R' + \lambda R\right) &= -\frac{\Theta''}{\Theta} \\ \downarrow \\ \frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) &= -\frac{\Theta''}{\Theta} \end{aligned}$$

LHS depend only on r while RHS depend only on θ , therefore both must be a constant

$$\begin{aligned} \frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) &= \beta \rightarrow \frac{1}{r}(rR')' + \lambda R = \frac{\beta R}{r^2} \\ -\frac{\Theta''}{\Theta} &= \beta \\ \downarrow \\ \begin{cases} \Theta'' = -\beta\Theta \\ R'' + \frac{1}{r}R' + R\left(\lambda - \frac{\beta}{r^2}\right) = 0 \end{cases} \end{aligned}$$

Solution to Θ :

$$\Theta(\theta) = A \cos(\sqrt{\beta}\theta) + B \sin(\sqrt{\beta}\theta)$$

$$\Theta(0) = A \rightarrow A = 0$$

$$\Theta\left(\frac{\pi}{3}\right) = B \sin\left(\sqrt{\beta}\frac{\pi}{3}\right) = 0 \rightarrow \sqrt{\beta}\frac{\pi}{3} = k\pi \rightarrow \beta = 9k^2, \quad k = 1, 2, 3\ldots$$

$$\Theta(\theta) = \sin(\sqrt{\beta}\theta)$$

Put the β into the equation for $R(r)$ and we get Bessel's differential equation:

$$R'' + \frac{1}{r}R' + \left(\lambda - \frac{9k^2}{r^2}\right)R = 0, \nu^2 = 9k^2 \rightarrow \nu = 3k$$

This has the solution

$$R(r) = a_k J_{3k}(\sqrt{\lambda}r) + b_k Y_{3k}(\sqrt{\lambda}r)$$

Since $R(r)$ has to be bounded, $b_k = 0$

$$R(r) = a_k J_{3k}(\sqrt{\lambda_k}r)$$

$$R(1) = 0 \rightarrow \lambda_k = \alpha_{3k,n}^2$$

The full solution is then

$$u(r, \theta) = a_k J_{3k}(a_{3k,n}r) \sin(3k\theta)$$

$$\lambda_{kn} = \alpha_{3k,n}^2, \quad k, n = 1, 2, 3\ldots$$

□

S.16

The spherical Bessel functions are defined as

$$j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}}(x)$$

Calculate j_0, j_1 , and j_2 by using

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin(x)$$

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}, \quad \nu = 0, 1, 2\ldots$$

Solution:

$$j_0(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{1}{2}}(x) = \sin(x)$$

$$j_1(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{3}{2}}(x) = \frac{\sin(x) - x \cos(x)}{x^2}$$

$$j_2(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{5}{2}}(x) = \sqrt{\frac{\pi}{2x}} (-x^{\frac{3}{2}}) \frac{d}{dx} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right)$$

$$\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} = \frac{1}{x^{\frac{3}{2}}} \sqrt{\frac{2x}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^2} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^3} \right)$$

$$\frac{d}{dx} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{x^4 \sin(x) - 3x^2 \sin(x) + 3x^3 \cos(x)}{x^6} \right)$$

$$j_2(x) = \left(\frac{(3 - x^2) \sin(x) - 3x \cos(x)}{x^3} \right)$$

□

S.18

Determine the general solution to

(a) $r^2 y'' + 2ry' = 0$

(b) $r^2 y'' + 2ry' - 2y = 0$

(c) $r^2 y'' + 2ry' + r^2 y = 0$

Solution:

The general solution to the operator

$$\mathcal{A} = y'' + \frac{2}{r} y' + \left(\lambda - \frac{l(l + \frac{1}{2})}{r^2} \right) y$$

is

$$\begin{cases} aj_l(\sqrt{\lambda}r) + by_l(\sqrt{\lambda}r), & \lambda > 0 \\ ar^l + br^{-l-1}, & \lambda = 0, l \neq -\frac{1}{2} \\ \frac{a+b\ln(r)}{\sqrt{r}}, & \lambda = 0, l = -\frac{1}{2} \end{cases}$$

(a)

$$r^2 y'' + 2ry' = 0 \rightarrow y'' + \frac{2}{r}y' = 0$$

Here we have $\lambda = 0, l = 0$, so the solution is:

$$y(r) = a + \frac{b}{r}$$

(b)

$$r^2 y'' + 2ry' - 2y = 0 \rightarrow y'' + \frac{2}{r}y' - \frac{2}{r^2}y = 0$$

$$\downarrow$$

$$y'' + \frac{2}{r}y' + \left(0 - \frac{2}{r^2}\right)y = 0$$

Here we have $\lambda = 0, l(l+1) = 2 \rightarrow l = 1$:

$$y(r) = ar + \frac{b}{r^2}$$

(c)

$$r^2 y'' + 2ry' + r^2 y = 0 \rightarrow y'' + \frac{2}{r}y' + y = 0$$

$$\downarrow$$

$$y'' + \frac{2}{r}y' + \left(1 - \frac{0 \cdot (0+1)}{r^2}\right)y = 0$$

Here we have $\lambda > 0$ and $l = 0$ (or $l = -1$)

$$y(r) = aj_0(r) + by_0(r)$$

□

S.20

Let Ω = unit sphere in $\mathbb{R}^3$,

$$\mathcal{A} = -\Delta$$

$$D_{\mathcal{A}} = \{u \in C^2(\Omega) \mid u = 0 \text{ on } \partial\Omega\}.$$

(a) Determine the radial (θ, ϕ independent) eigenfunctions and eigenvalues.

(b) Expand the function $1(x, y, z) \equiv 1$ using eigenfunctions to $\mathcal{A}$.

Solution:

The radial component of the Laplacian in spherical coordinates is

$$\Delta_r = u'' + \frac{2}{r}u'$$

So we want to solve

$$\mathcal{A}u = \lambda u \rightarrow -\Delta_r u = \lambda u$$

↓

$$u'' + \frac{2}{r}u' + \lambda u = 0$$

↓

$$u'' + \frac{2}{r}u' + \left(\lambda - \frac{l(l+1)}{r^2}\right)u = 0$$

Here we have $\lambda > 0$ and $l = 0$ and we use the general solution to this and get

$$u(r) = aj_0(\sqrt{\lambda}r) + by_0(\sqrt{\lambda}r)$$

Since $u(r)$ has to be bounded at $r = 0$, the $y_0(\sqrt{\lambda}r)$ -term has to disappear, so $b = 0$ and our final solution is

$$u(r) = aj_0(\sqrt{\lambda}r)$$

The eigenvalues can be found from

$$u(1) = 0 \rightarrow aj_0(\sqrt{\lambda}) = 0$$

which means $\sqrt{\lambda}$ has to make $j_0 = 0$. But $j_0(\sqrt{\lambda}r) = \frac{\sin(\sqrt{\lambda}r)}{r}$ so that gives

$$\sqrt{\lambda} = k\pi$$

↓

$$\lambda_k = k^2\pi^2$$

and the eigenfunctions are

$$\varphi_k(r) = \frac{\sin(k\pi r)}{r}$$

(b)

$$1 = \sum_{k=1}^{\infty} c_k \varphi_k(r)$$

$$\begin{aligned}
 c_k &= \frac{(\varphi_k \mid 1)}{(\varphi_k \mid \varphi_k)} = \frac{\int_0^1 r \sin(k\pi r) \, dr}{\int_0^1 \sin^2(k\pi r) \, dr} = \frac{\left[\frac{-r \cos(k\pi r)}{k\pi} \right]_0^1 + \underbrace{\int_0^1 \frac{\cos(k\pi r)}{k\pi} \, dr}_{=0}}{\int_0^1 \left(\frac{1}{2} - \underbrace{\frac{\cos(2k\pi r)}{2}}_{\int=0} \right) \, dr} = \\
 &= -2 \frac{\cos(k\pi)}{k\pi} \rightarrow c_k = \frac{2}{k\pi} (-1)^{k+1}
 \end{aligned}$$

□